The Linked Data Benchmark Council (LDBC): Driving competition and collaboration in the graph data management space

Gábor Szárnyas

TPCTC | 2023-08-28 | Vancouver

Co-authors: Brad Bebee, Altan Birler, Alin Deutsch, George Fletcher, Henry A. Gabb, Denise Gosnell, Alastair Green, Zhihui Guo, Keith W. Hare, Jan Hidders, Alexandru Iosup, Atanas Kiryakov, Tomas Kovatchev, Xinsheng Li, Leonid Libkin, Heng Lin, Xiaojian Luo, Arnau Prat-Pérez, David Püroja, Shipeng Qi, Oskar van Rest, Benjamin A. Steer, Dávid Szakállas, Bing Tong, Jack Waudby, Mingxi Wu, Bin Yang, Wenyuan Yu, Chen Zhang, Jason Zhang, Yan Zhou, Peter Boncz

LDBC: Linked Data Benchmark Council

Non-profit company

Mission: Accelerate progress in graph data management

Designs graph benchmarks & governs their use

Fosters collaboration between researchers & practitioners

ldbcouncil.org

github.com/ldbc

Sponsors

ANT GROUP

火山引擎 Beijing Volcano (ByteDance)

ORACLE Labs

Companies and Research Institutes

aws

intel.

neo4j

TigerGraph

NebulaGraph

JCC Consulting, Inc.

KATANA GRAPH

MEM GRAPH

ontotext

Fabarta

创邻科技 CREATE LINK

pometry

ULTIPA™

*Sparsity

海致星图 StarGraph

達摩院 ALIBABA DAMO ACADEMY

RelationalAI

FORTH

之江实验室 ZHEJIANG LAB

ArangoDB

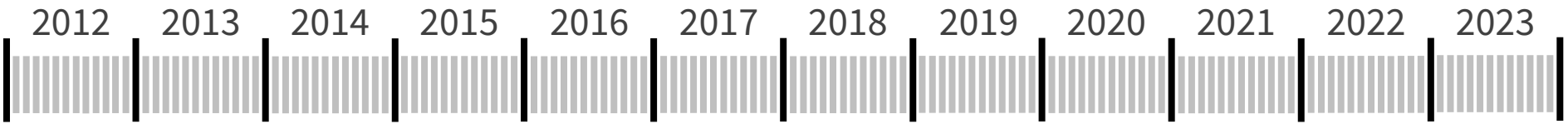
Birkbeck UNIVERSITY OF LONDON

My involvement in LDBC

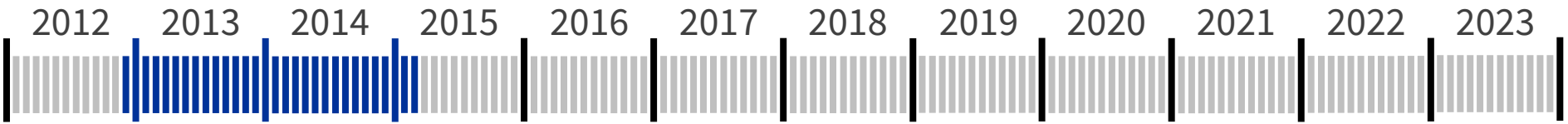
2017	Joined a benchmark task force
2020	Started working at CWI in Amsterdam *(Database Architectures group)*
Tasks	Benchmarks and their auditing process Organizational restructuring Running board and community meetings

LDBC’s history

LDBC timeline

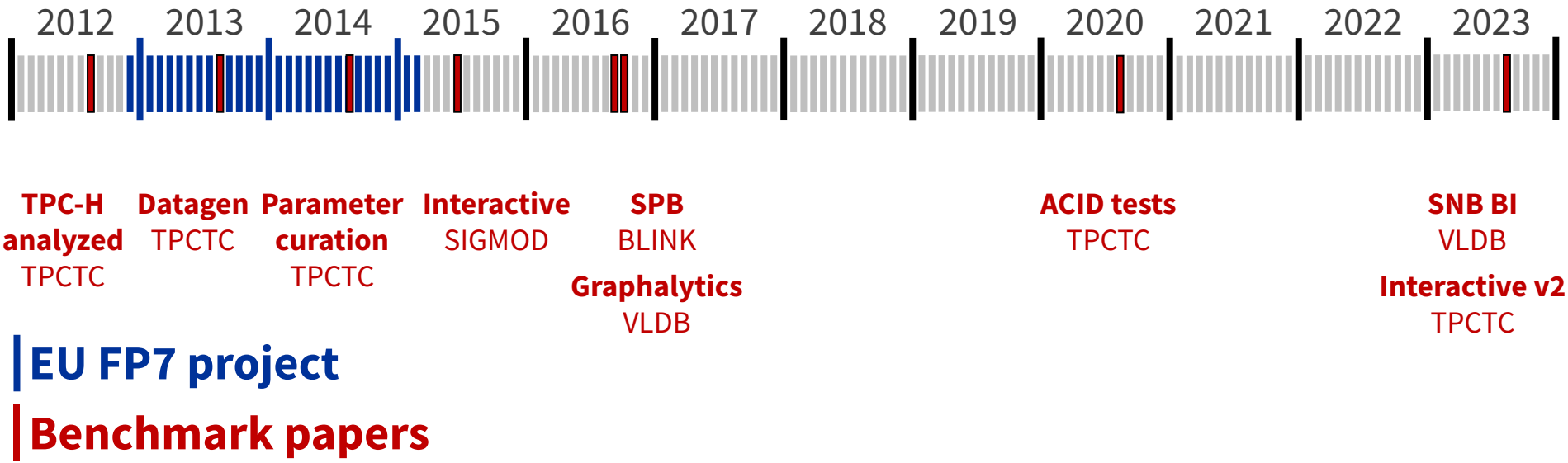


LDBC timeline

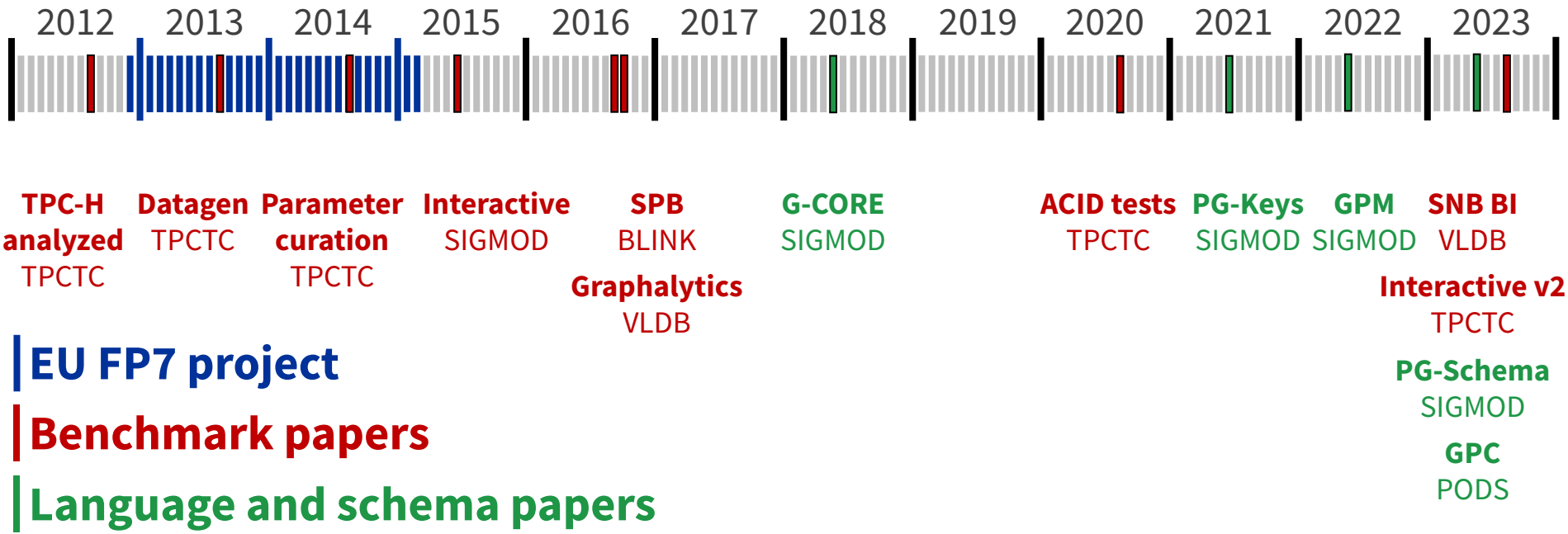


EU FP7 project

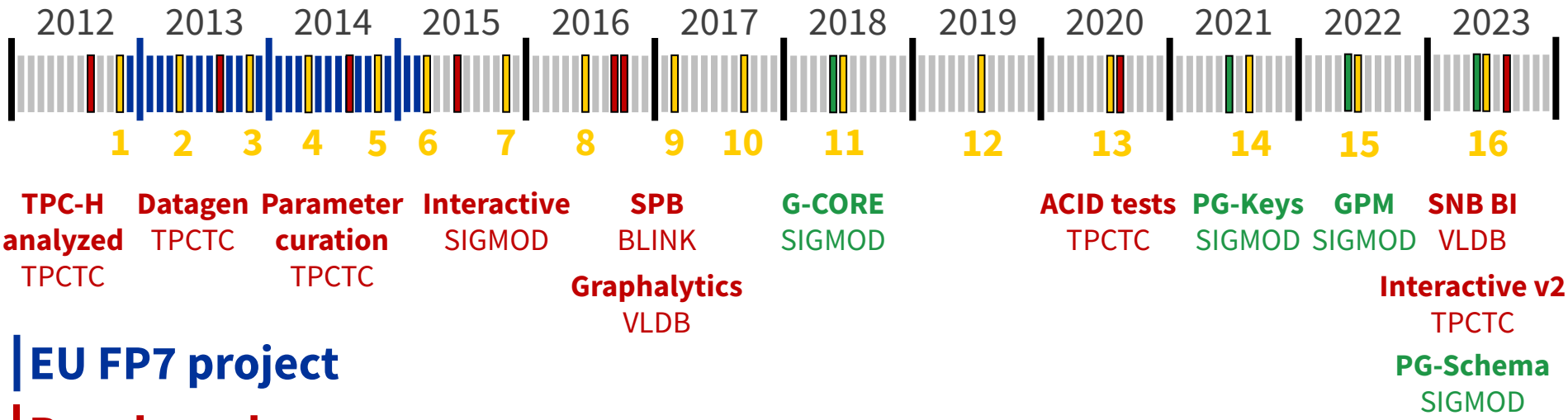
LDBC timeline



LDBC timeline



LDBC timeline



EU FP7 project

Benchmark papers

Language and schema papers

Technical User Community meetings

Benchmark overview

Similarities to TPC benchmarks

application-level benchmarks	scale factors: SF30 = 30GiB CSV	few dozen query templates
third-party auditors	FDRs with metrics, e.g. throughput@SF	benchmark approval and renewal

Similarities to TPC benchmarks

application-level benchmarks	scale factors: SF30 = 30GiB CSV	few dozen query templates
third-party auditors	FDRs with metrics, e.g. throughput@SF	benchmark approval and renewal

Similarities to TPC benchmarks

application-level benchmarks	scale factors: SF30 = 30GiB CSV	few dozen query templates
third-party auditors	FDRs with metrics, e.g. throughput@SF	benchmark approval and renewal

Similarities to TPC benchmarks

application-level benchmarks	scale factors: SF30 = 30GiB CSV	few dozen query templates
third-party auditors	FDRs with metrics, e.g. throughput@SF	benchmark approval and renewal

Similarities to TPC benchmarks

application-level benchmarks	scale factors: SF30 = 30GiB CSV	few dozen query templates
third-party auditors	FDRs with metrics, e.g. throughput@SF	benchmark approval and renewal

Similarities to TPC benchmarks

application-level benchmarks	scale factors: SF30 = 30GiB CSV	few dozen query templates
third-party auditors	FDRs with metrics, e.g. throughput@SF	benchmark approval and renewal

Similarities to TPC benchmarks

application-level benchmarks	scale factors: SF30 = 30GiB CSV	few dozen query templates
third-party auditors	FDRs with metrics, e.g. throughput@SF	benchmark approval and renewal

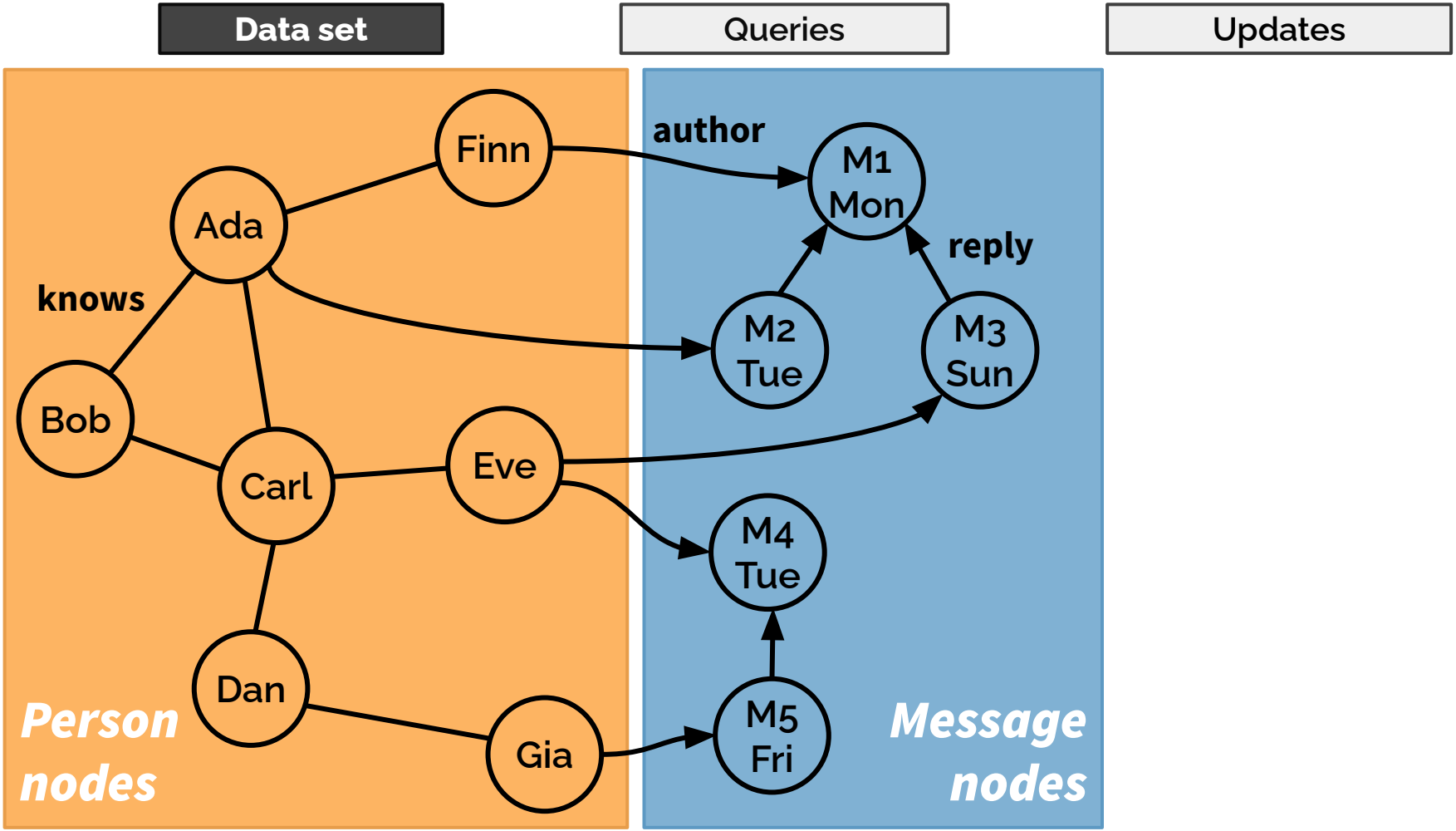
The Social Network Benchmark (SNB) suite

Data set and queries

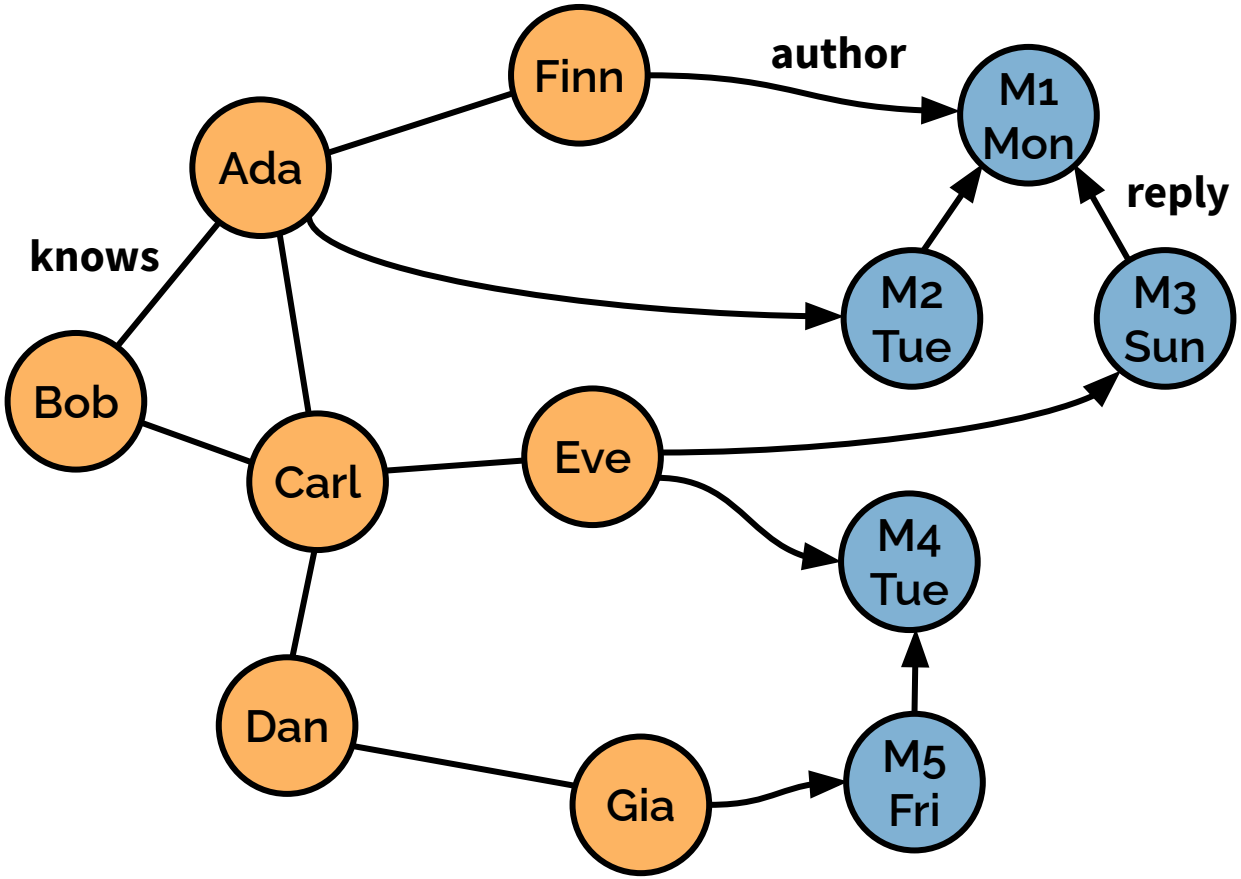
Data set

Queries

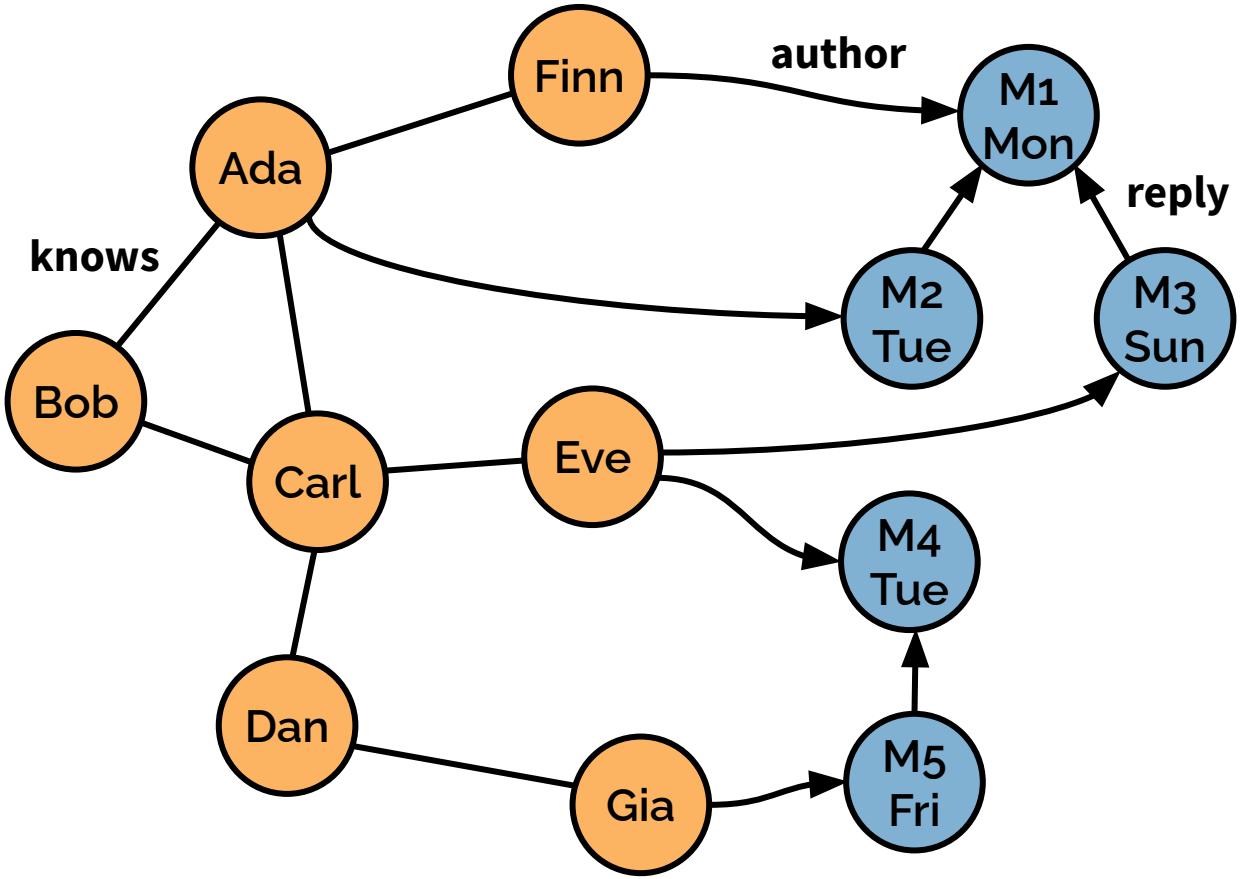
Updates


Data set
Queries
Updates
Finn
Ada
knows
Bob
Carl
Eve
Dan
Gia
author
M1
Mon
reply
M2
Tue
M3
Sun
M4
Tue
M5
Fri
Person nodes
Message nodes

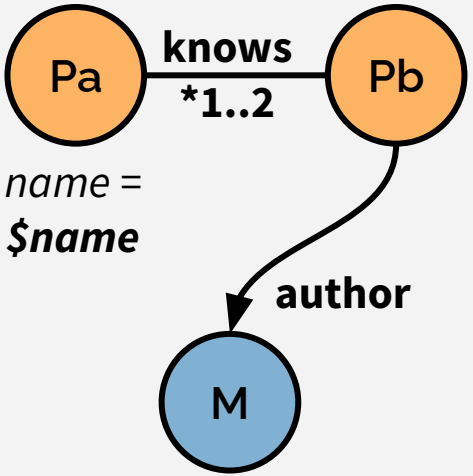
Data set | **Queries** | Updates



Data set | **Queries** | Updates

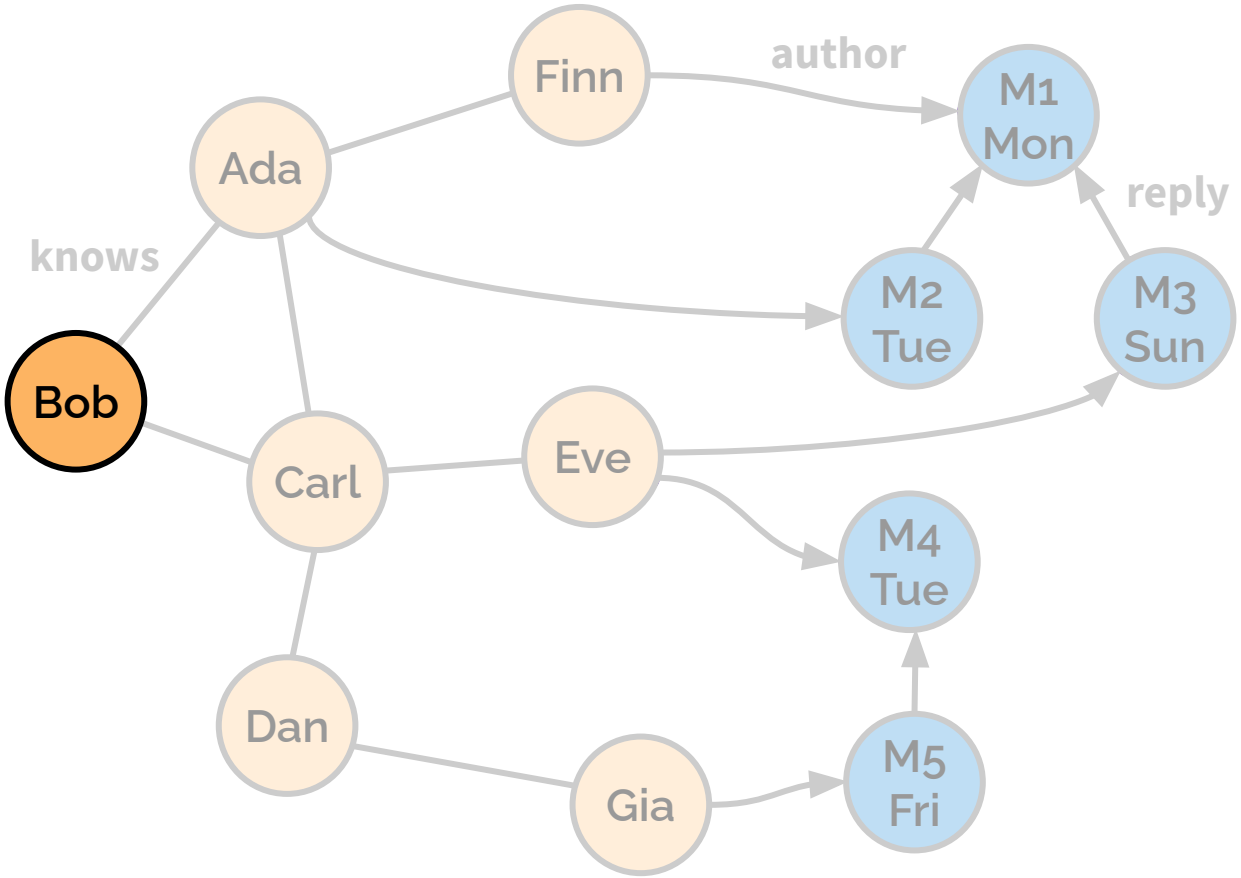


Q9(*$name*, *$day*)

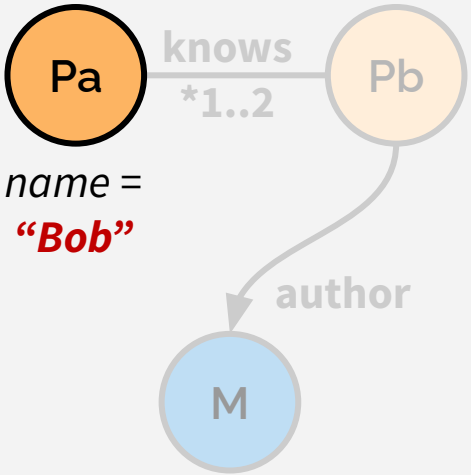


creation date < ***$day***

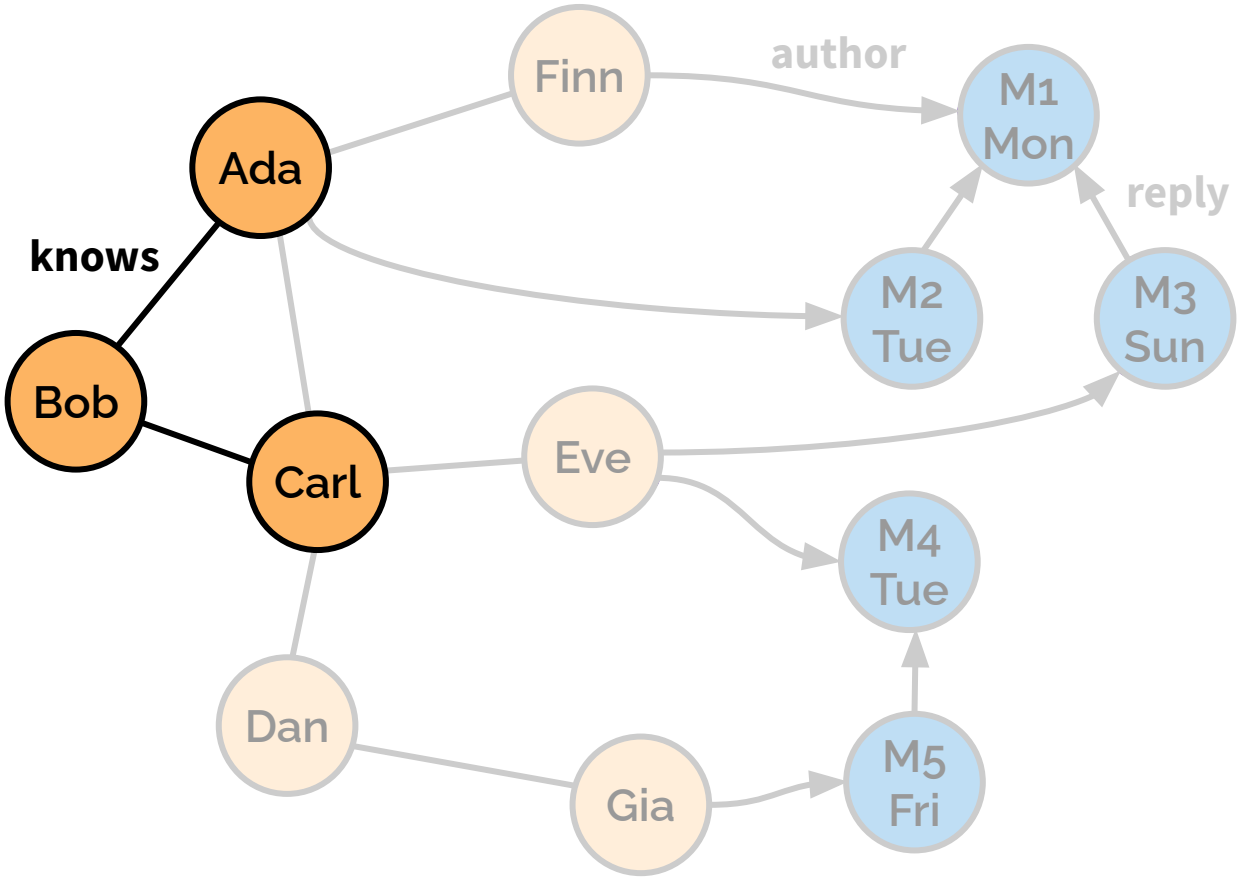
Data set | **Queries** | Updates



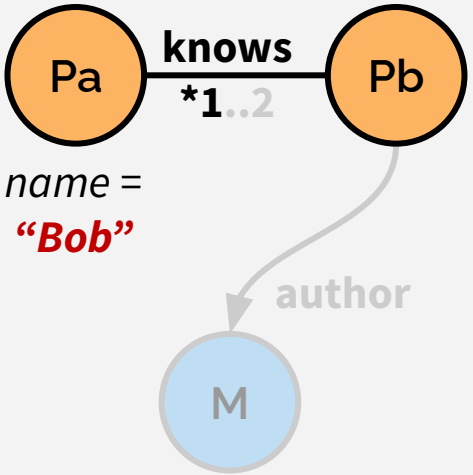
Q9(*"Bob"*, *"Sat"*)



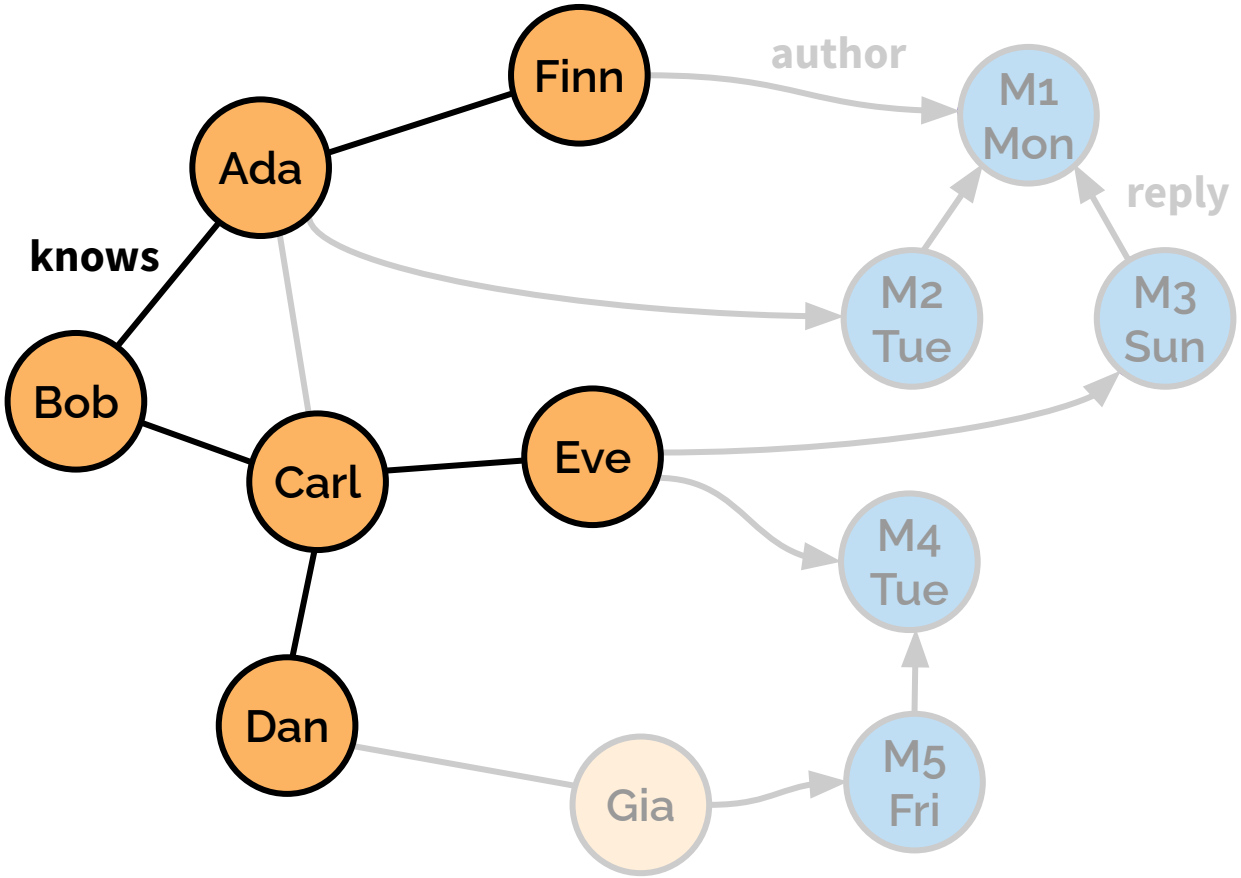
Data set | **Queries** | Updates



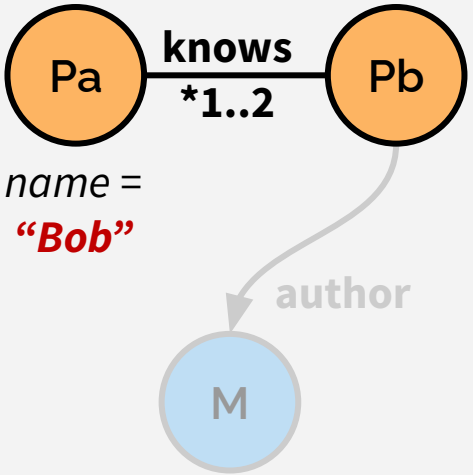
Q9(*"Bob"*, *"Sat"*)



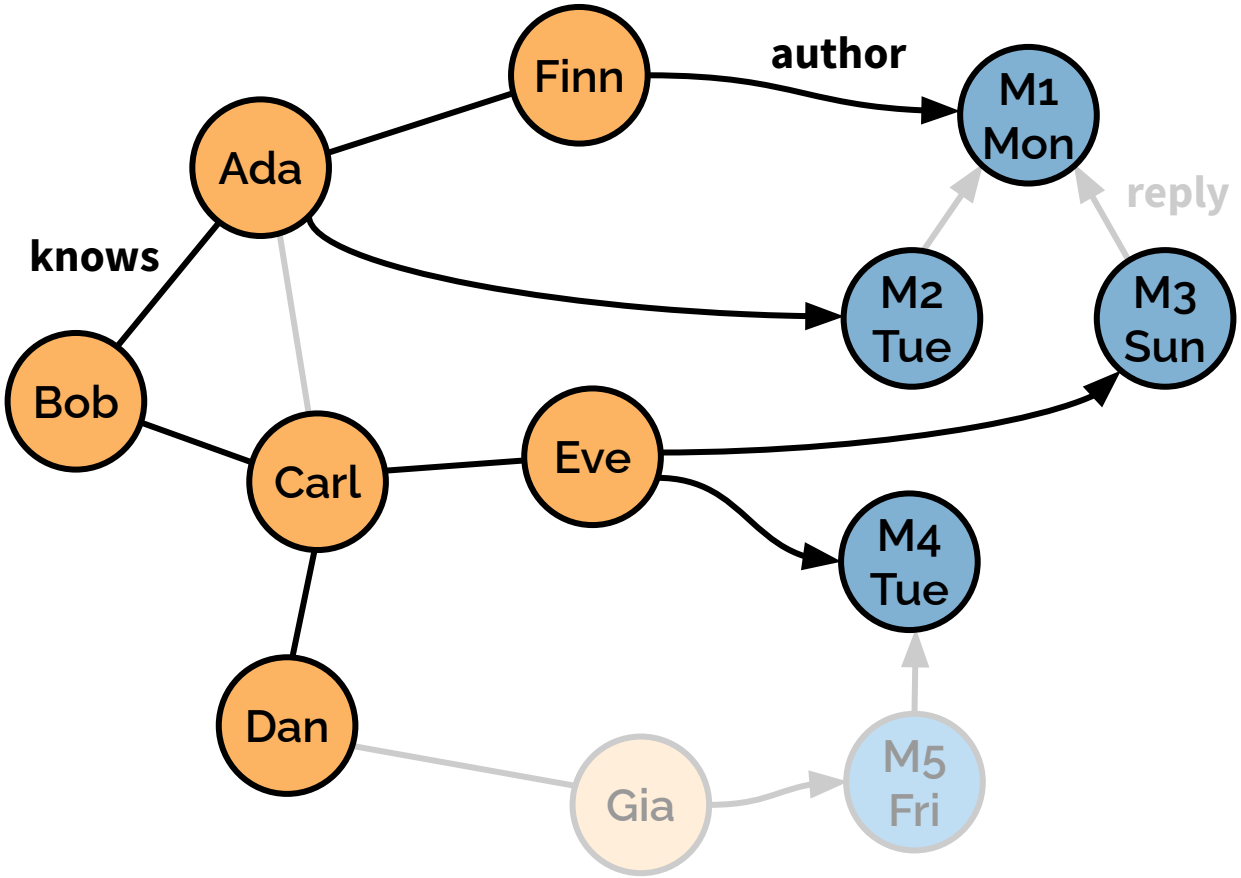
Data set | **Queries** | Updates



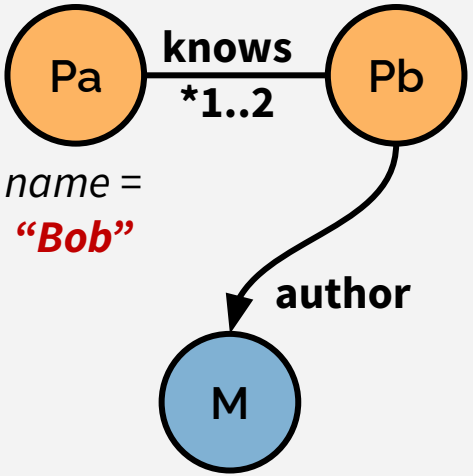
Q9(*"Bob"*, *"Sat"*)



Data set | **Queries** | Updates



Q9(*"Bob"*, *"Sat"*)

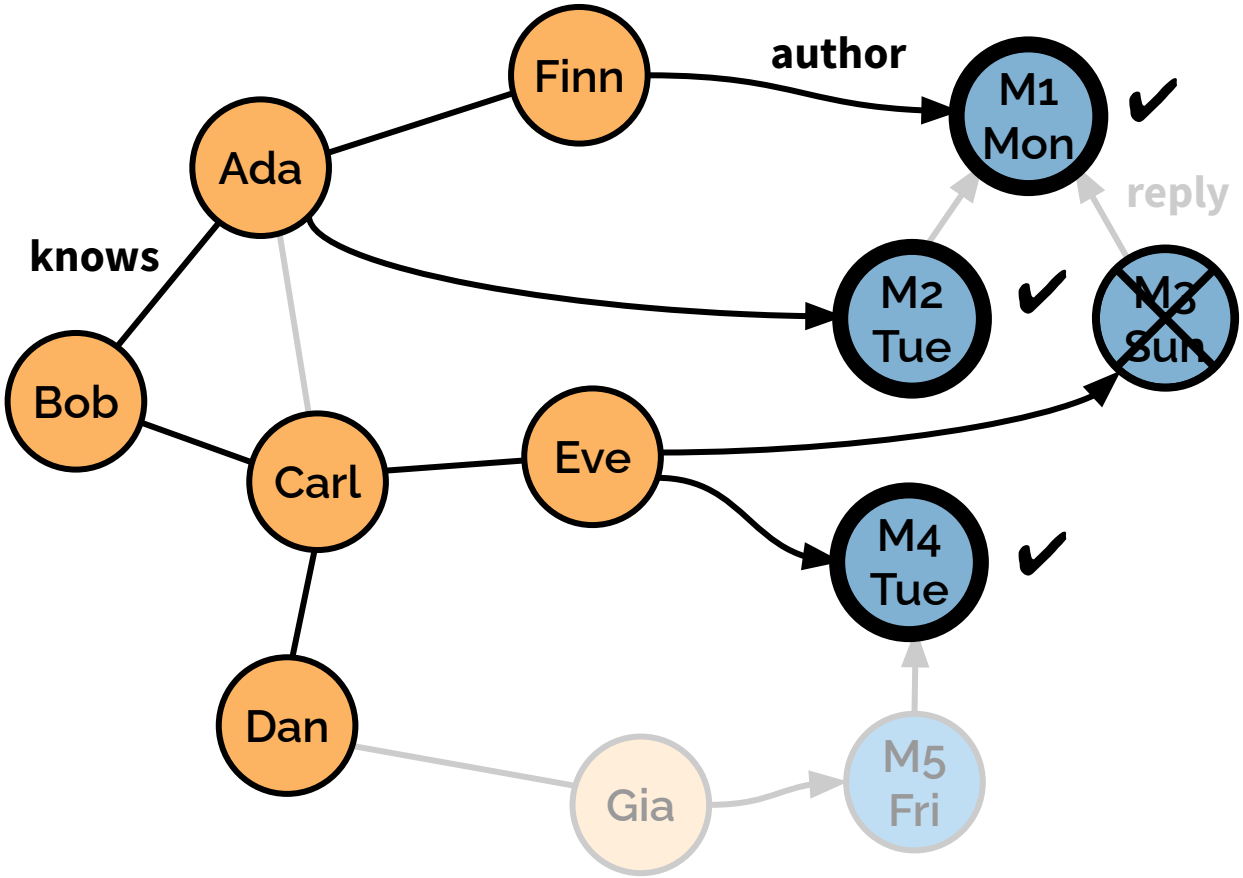


creation date < "Sat"

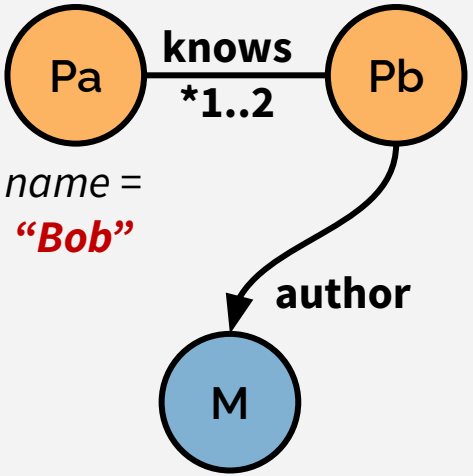
Data set

Queries

Updates

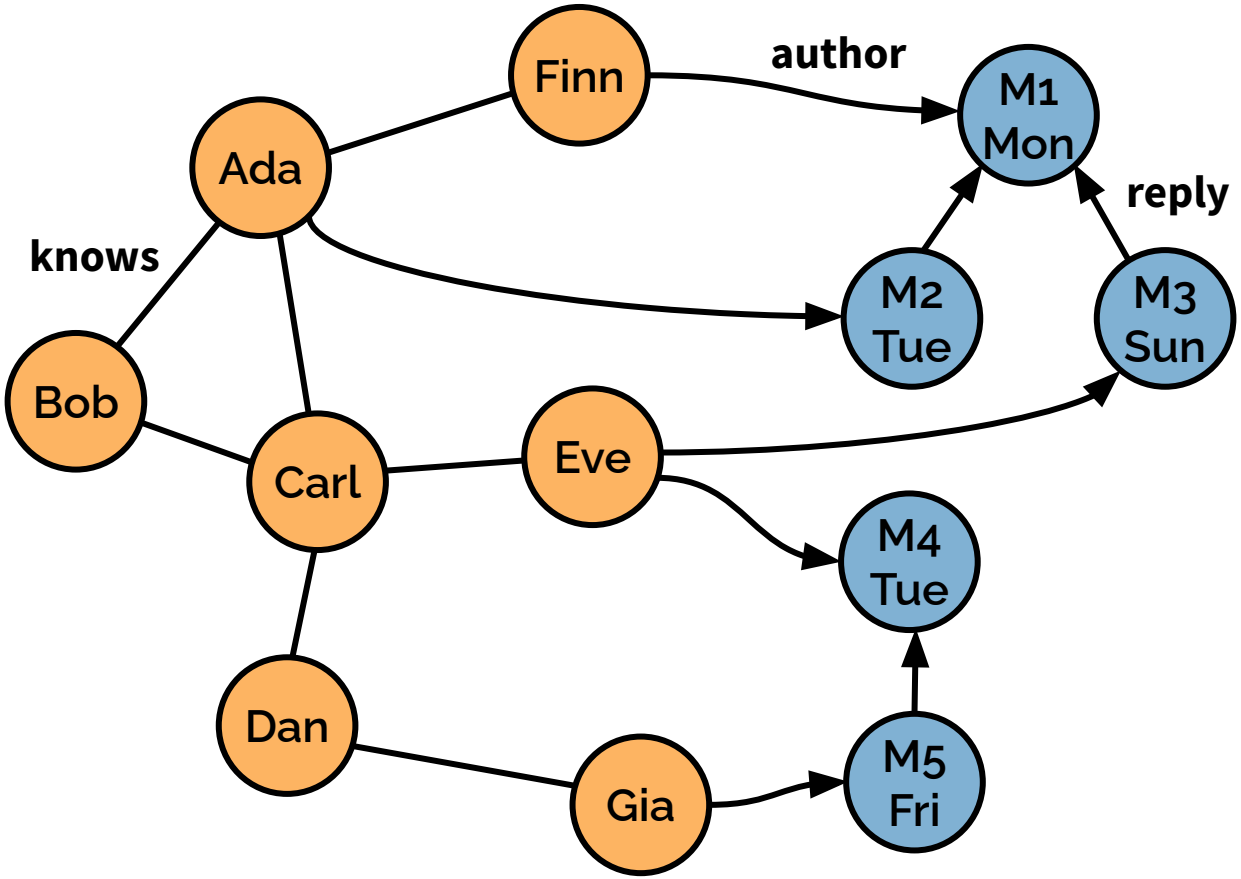


Q9(*"Bob"*, *"Sat"*)

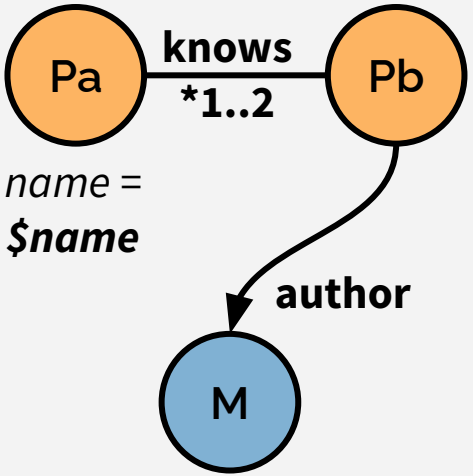


creation date < ***"Sat"***

Data set | **Queries** | Updates

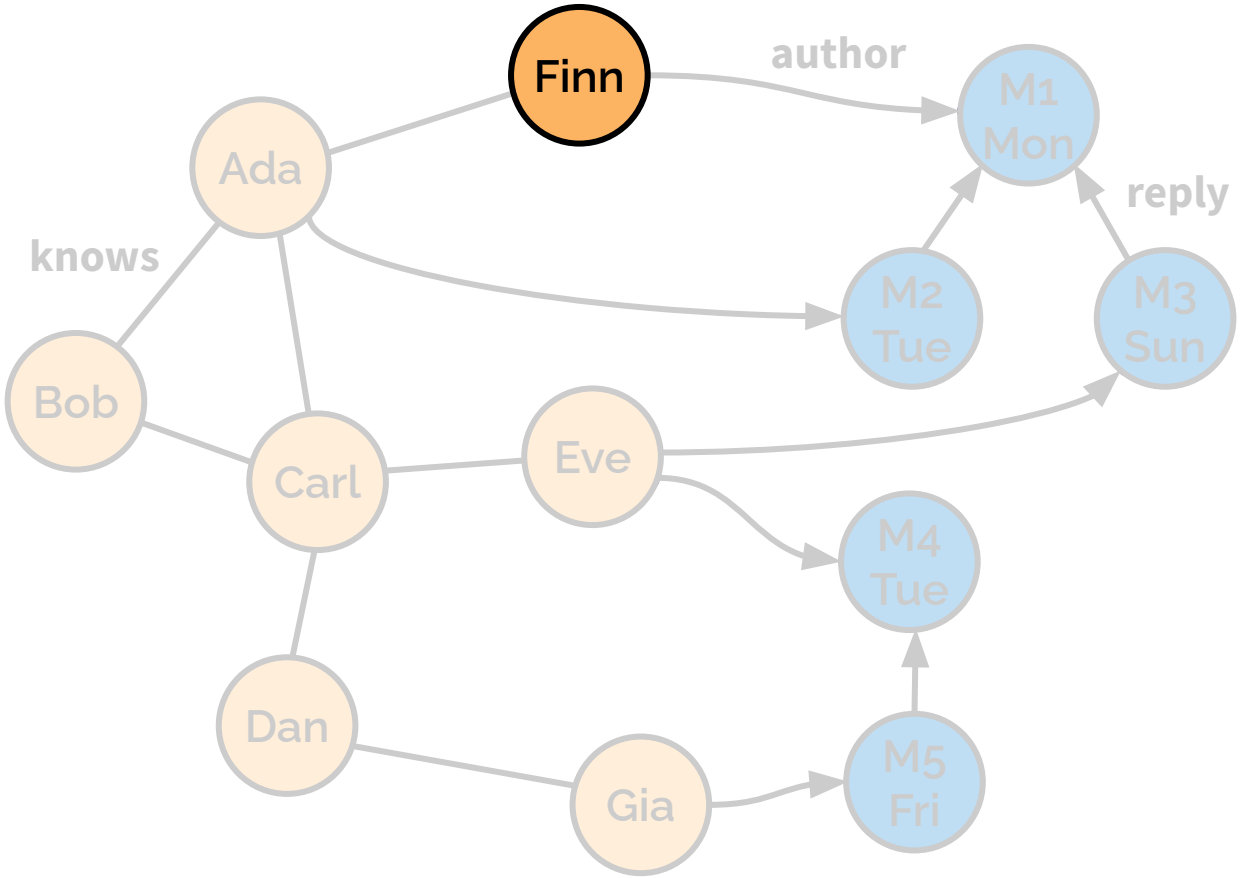


Q9(*$name*, *$day*)

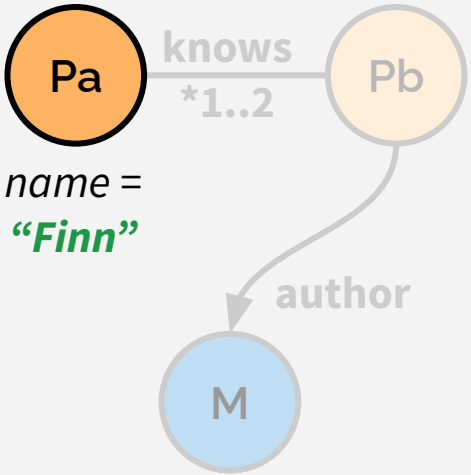


creation date < ***$day***

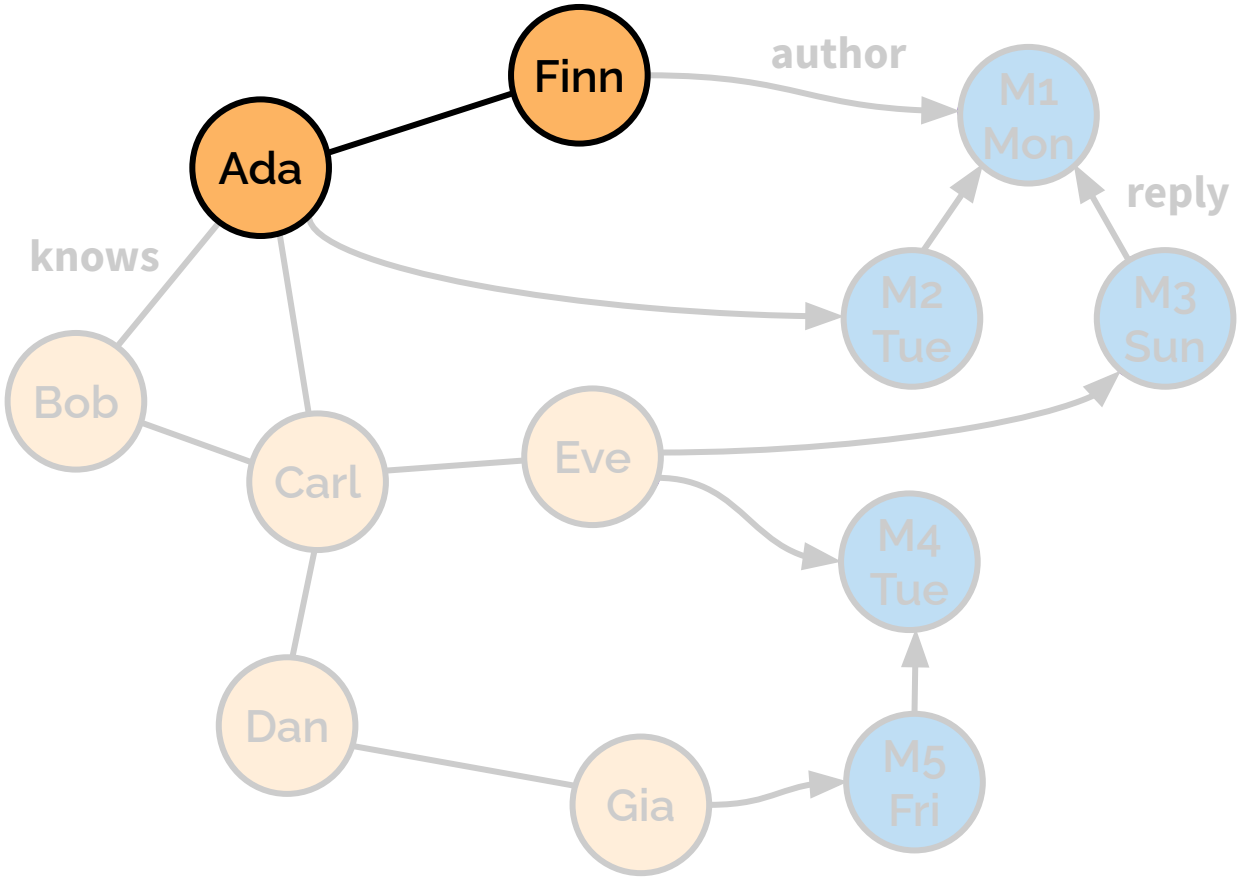
Data set | **Queries** | Updates



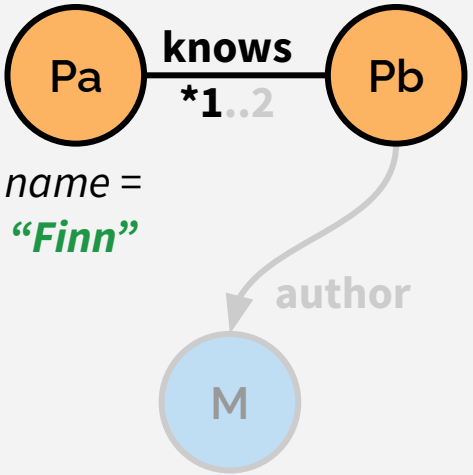
Q9(*"Finn"*, *"Wed"*)



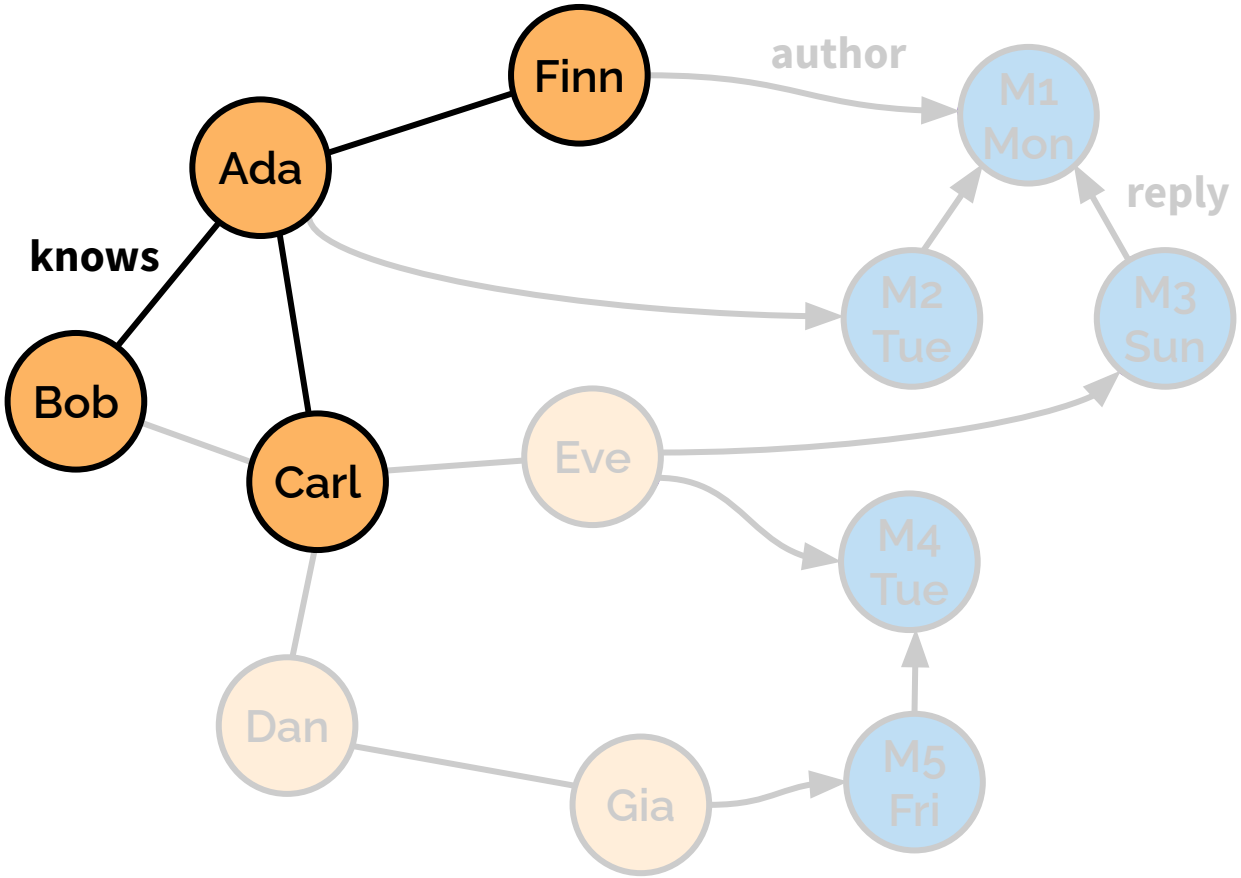
Data set | **Queries** | Updates



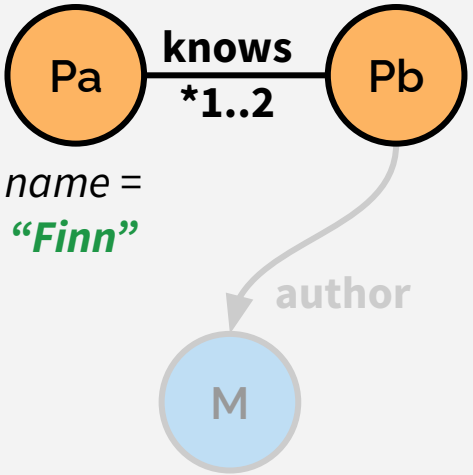
Q9(*"Finn"*, *"Wed"*)



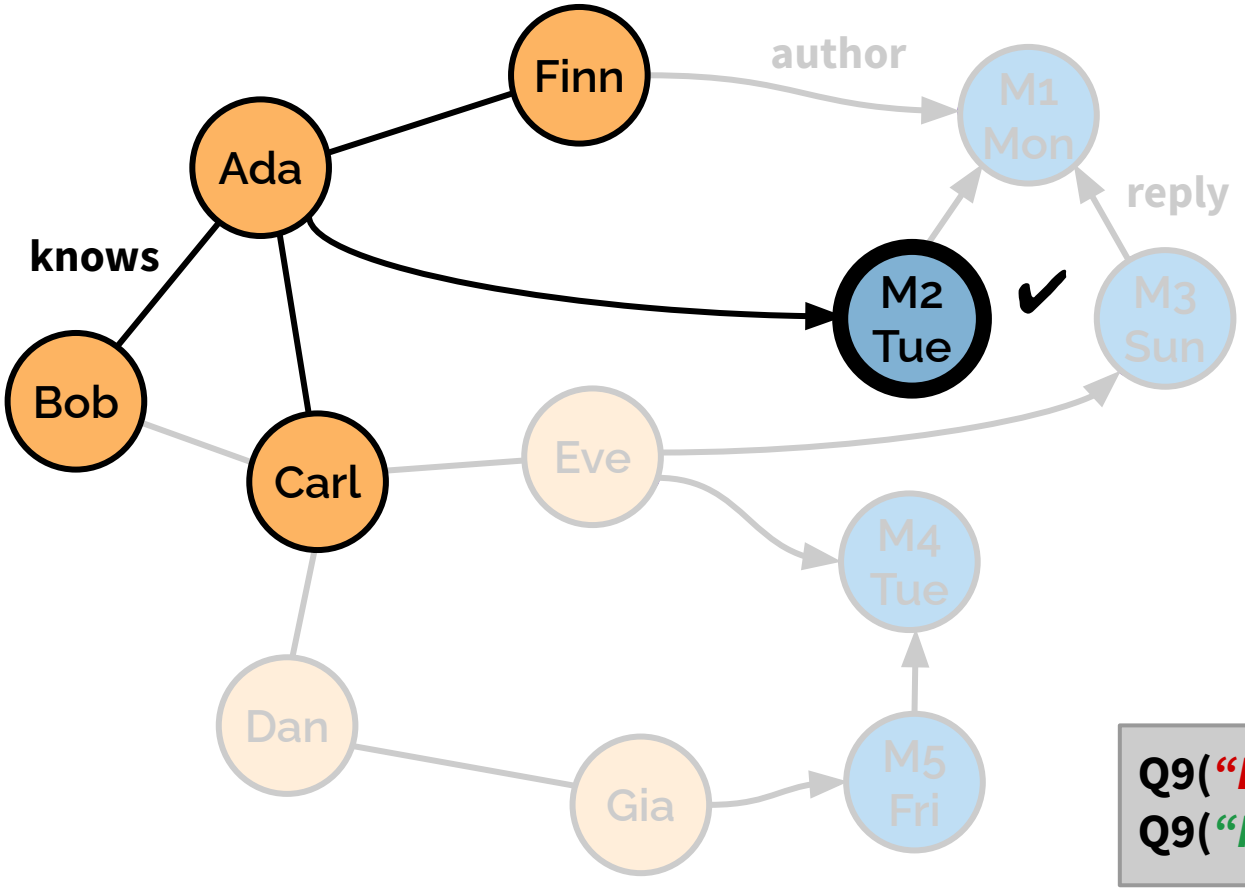
Data set | **Queries** | Updates



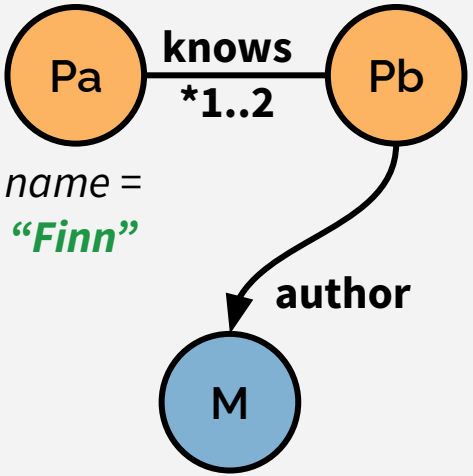
Q9(*"Finn"*, *"Wed"*)



Data set | **Queries** | Updates



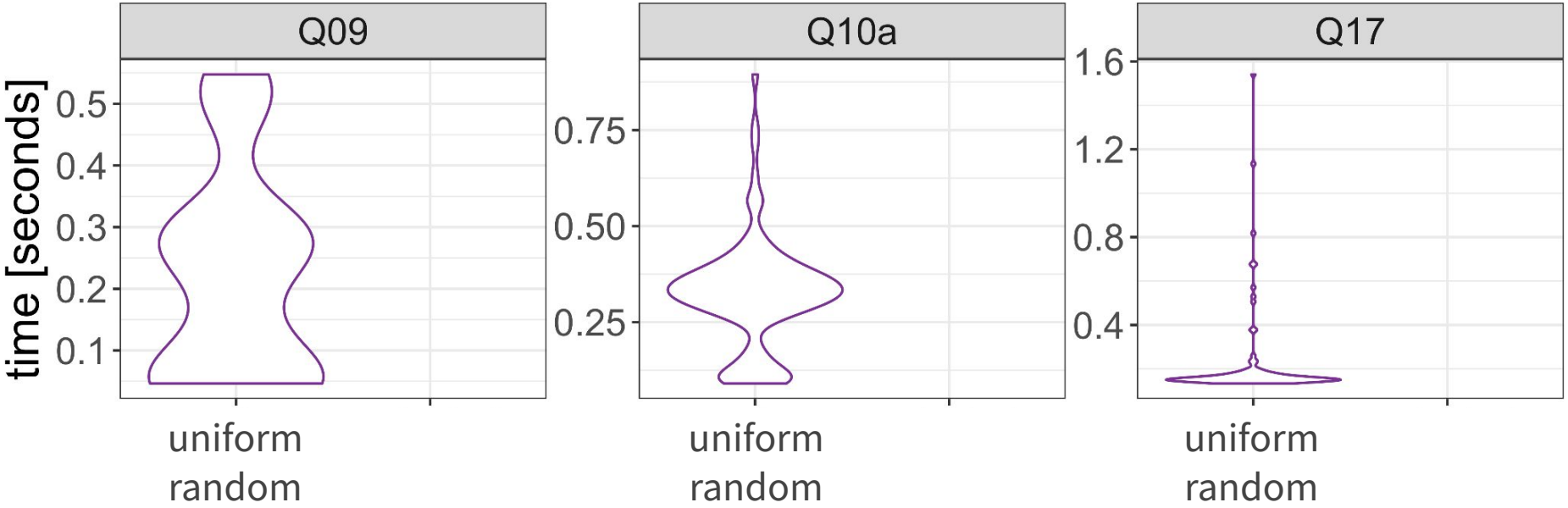
Q9(*"Finn"*, *"Wed"*)



Q9(*"Bob"*, *"Sat"*): 10 nodes
Q9(*"Finn"*, *"Wed"*): 5 nodes

Parameter selection

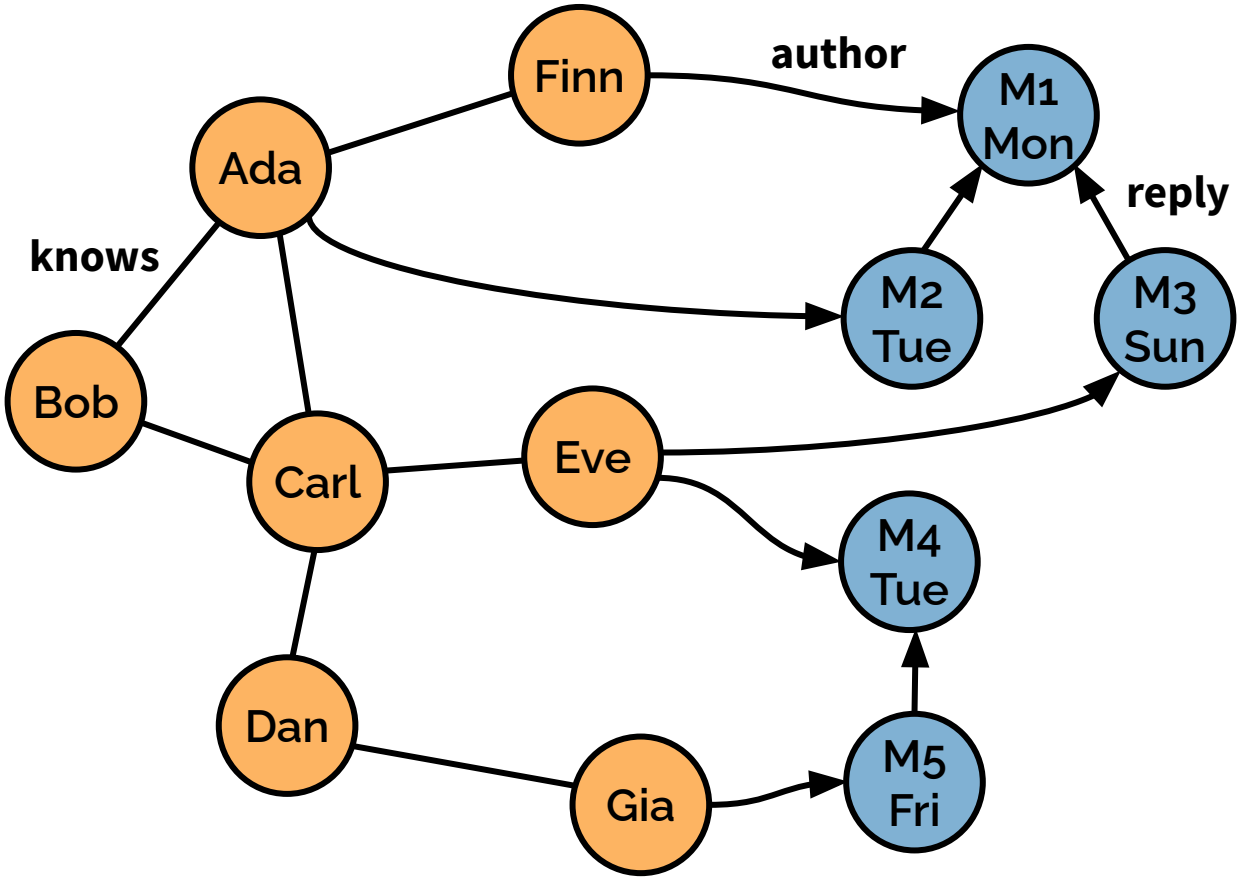
- ***Uniform random parameters*** → unstable distributions



Parameter curation

A. Gubichev, P. Boncz (TPCTC 2014)

Data set | **Queries** | Updates



Factor tables

numFriendsOfFriends

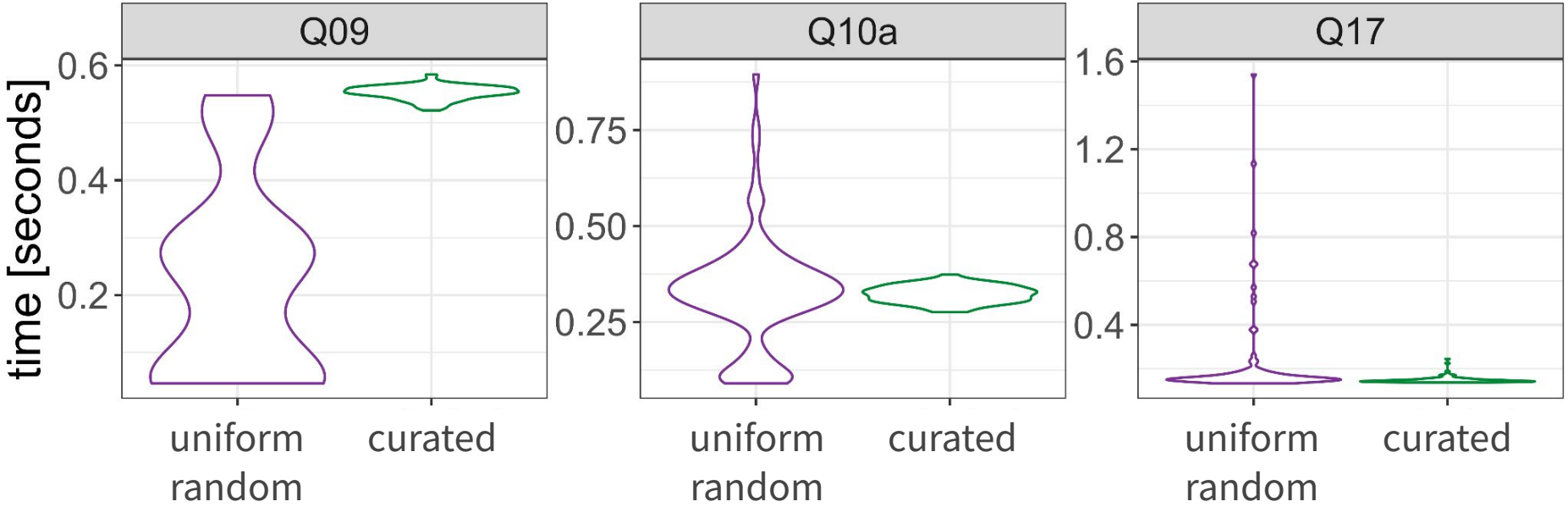
name	#1-hop	#2-hop
Bob	2	3
Carl	4	2
Ada	3	2
…		

numMessagesPerDay

day	#
Mon	1
Tue	2
…	

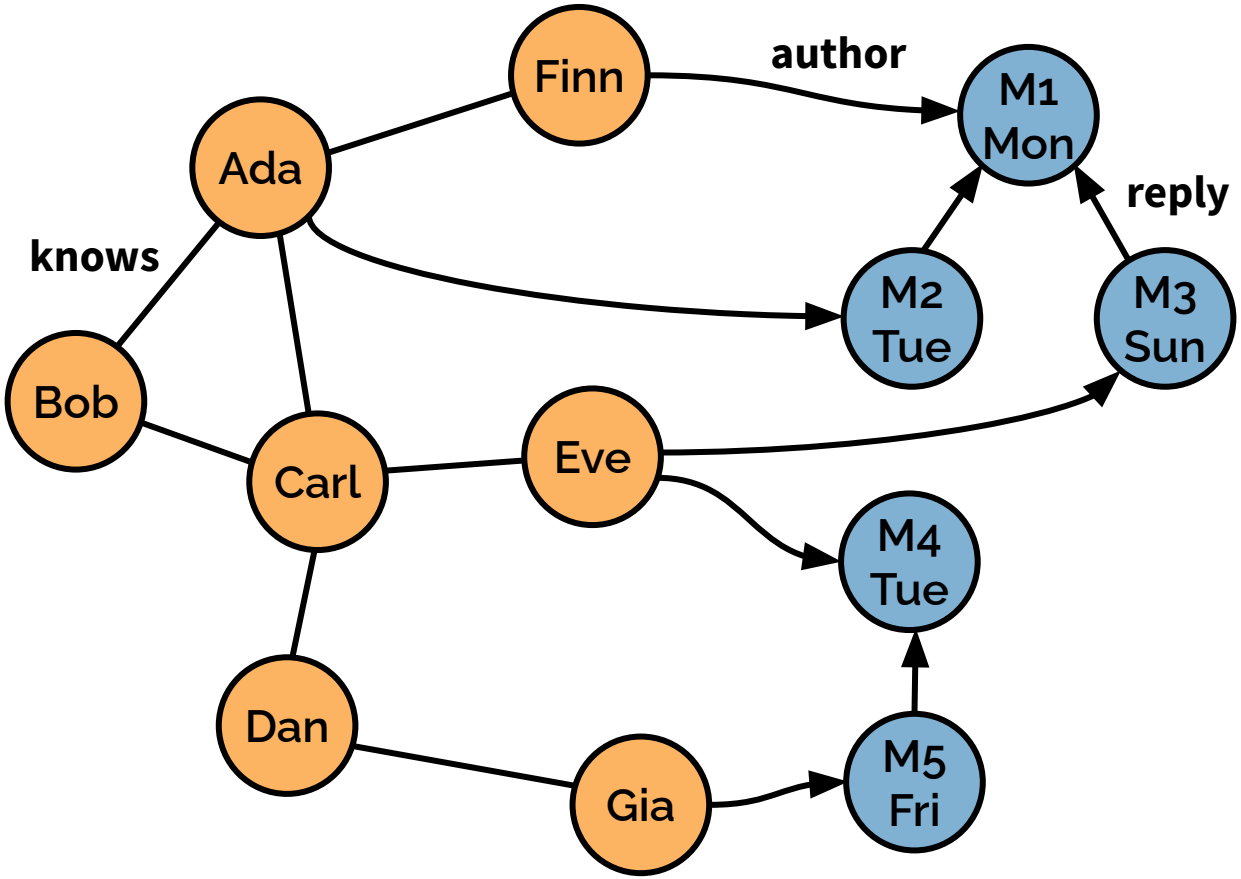
Parameter selection

- ***Uniform random parameters*** → unstable distributions
- ***Curated parameters*** → tighter distributions, closer to bell curves



Updates

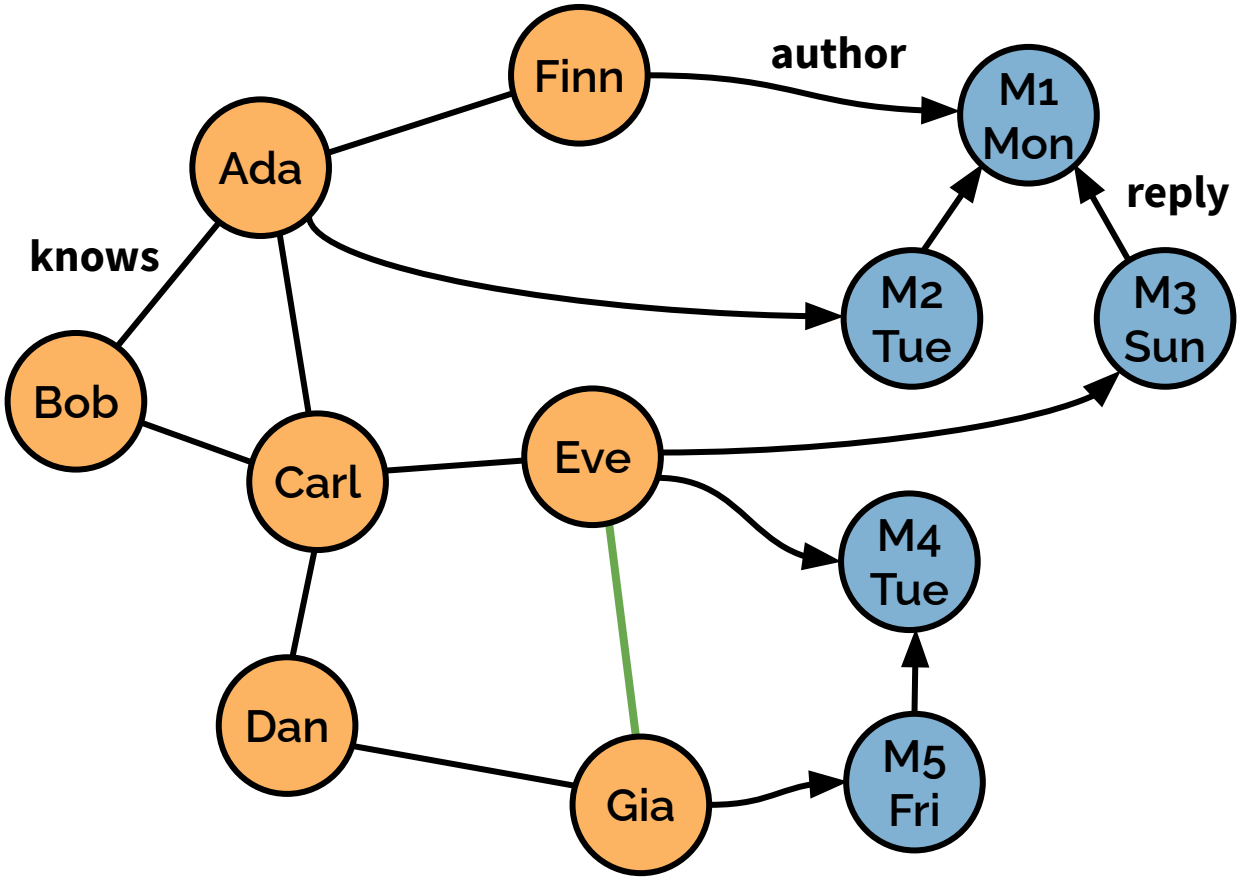
Data set | Queries | **Updates**



Data set

Queries

Updates



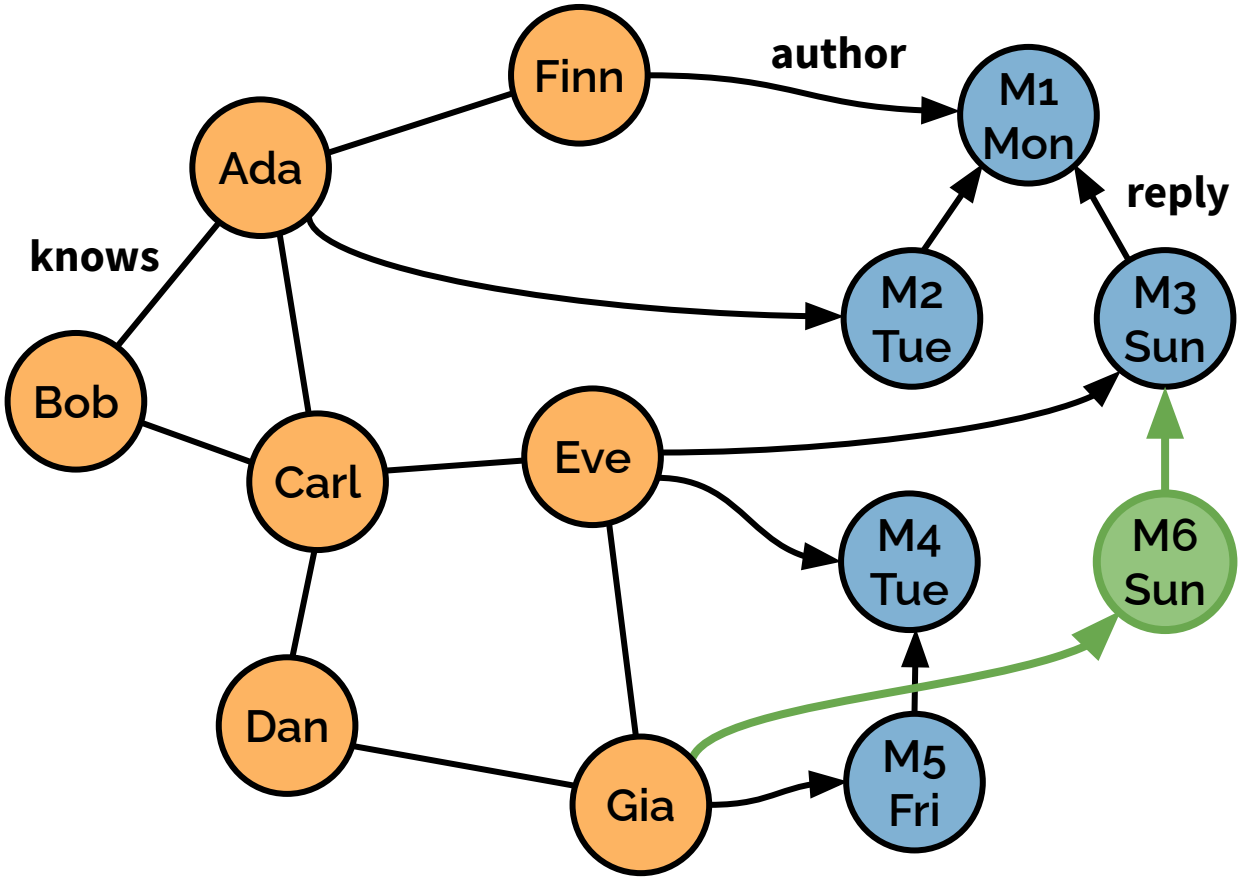
Updates

+ knows(“Eve”, “Gia”)

Data set

Queries

Updates



Updates

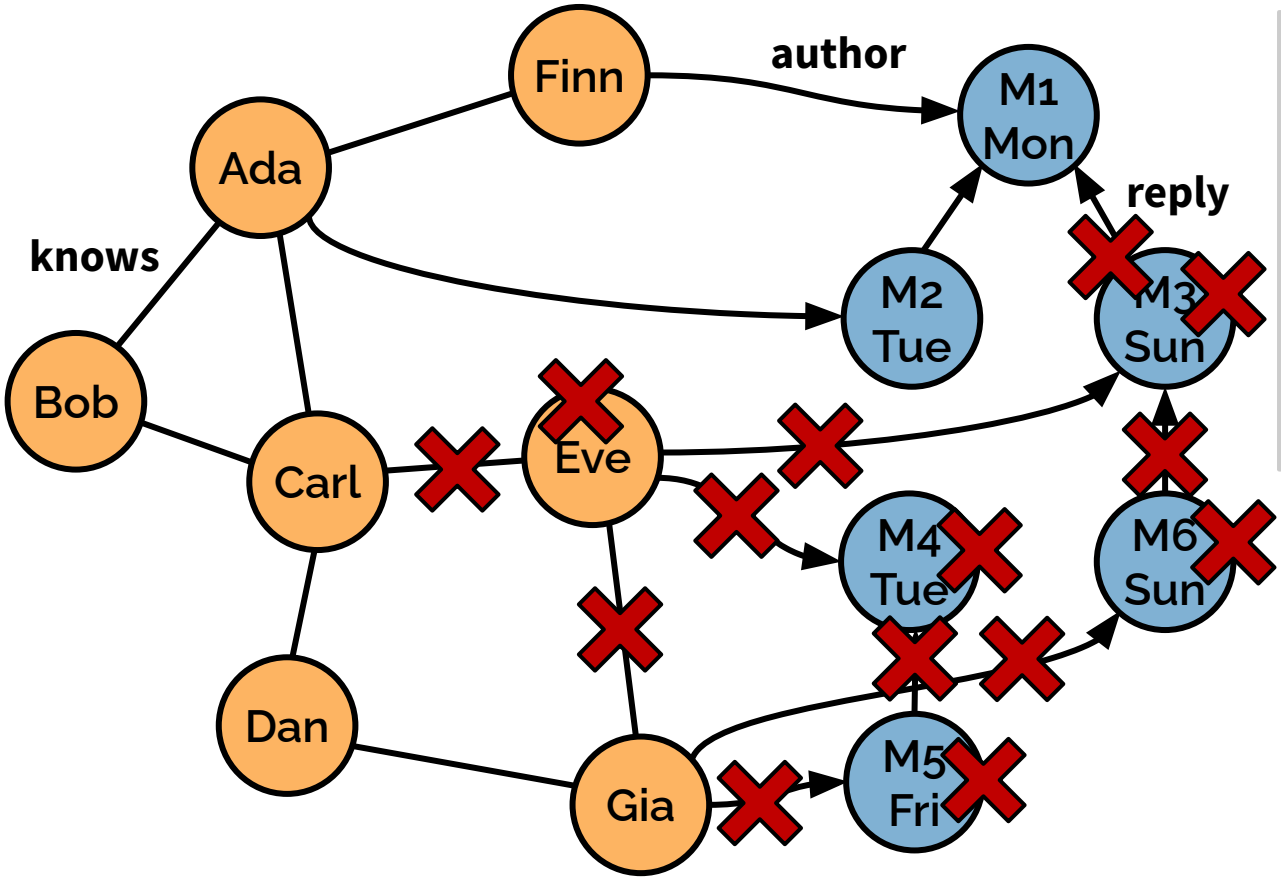
+ knows("Eve", "Gia")

+ Comment("Gia", "M3")

Data set

Queries

Updates



Updates

+ knows("Eve", "Gia")

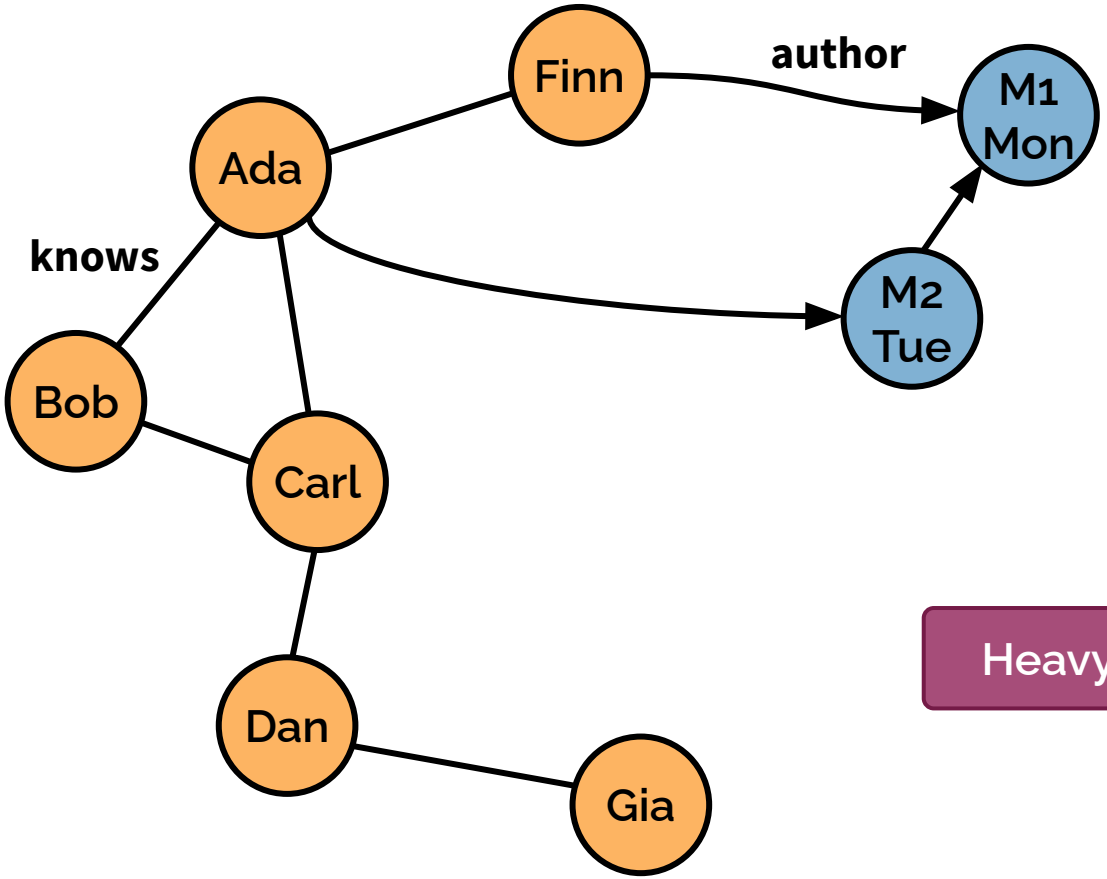
+ Comment("Gia", "M3")

– Person("Eve")

Data set

Queries

Updates



Updates

+ knows("Eve", "Gia")

+ Comment("Gia", "M3")

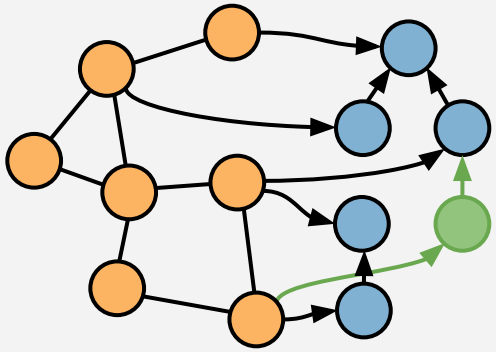
– Person("Eve")

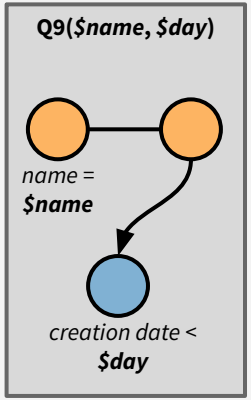
Heavy-hitting operation!

SNB workloads

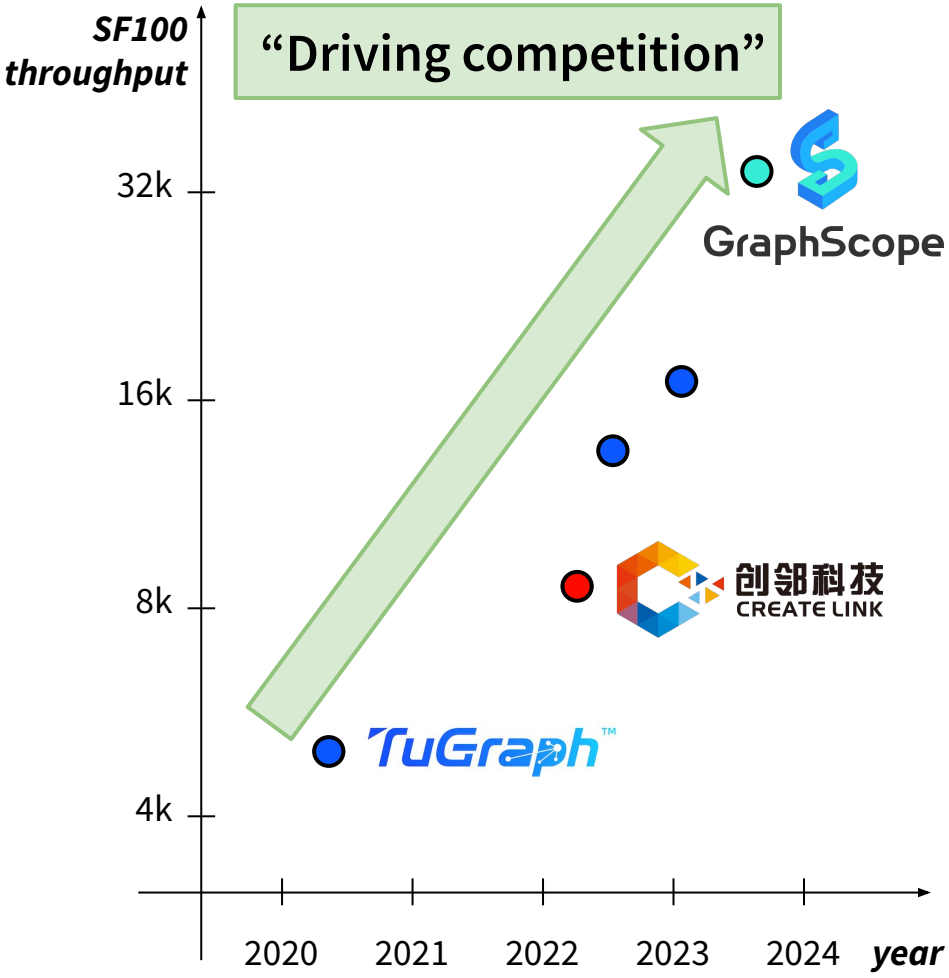
- OLTP: Interactive
- OLAP: Business Intelligence

SNB Interactive v1 (2015)

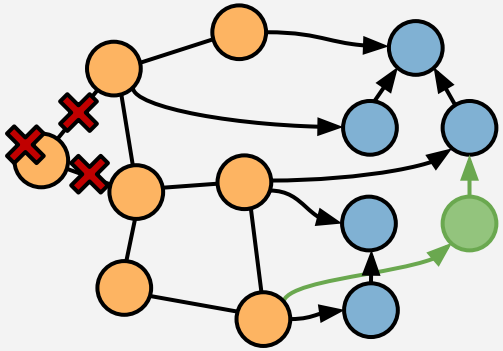


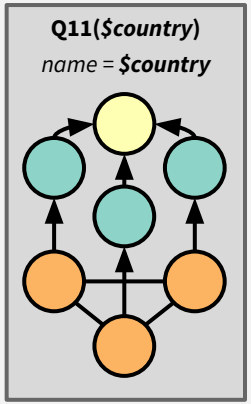


- Queries start in 1–2 person nodes
- 14 complex reads, 7 short reads
- 8 insert operations run concurrently
- Goal: High throughput (ops/s)



SNB Business Intelligence (2022)





Queries touch on large portions of the data

20 complex read queries, insert & delete ops

Both bulk and concurrent updates allowed

Goal: High throughput & low query runtimes

Audited results

TigerGraph + AMD

Results for 100GB, 1TB, and 10TB

Scores for 10TB:

- Power@SF: 89,444
- Throughput@SF: 30,990

More results expected in late 2023

Financial Benchmark (2023)

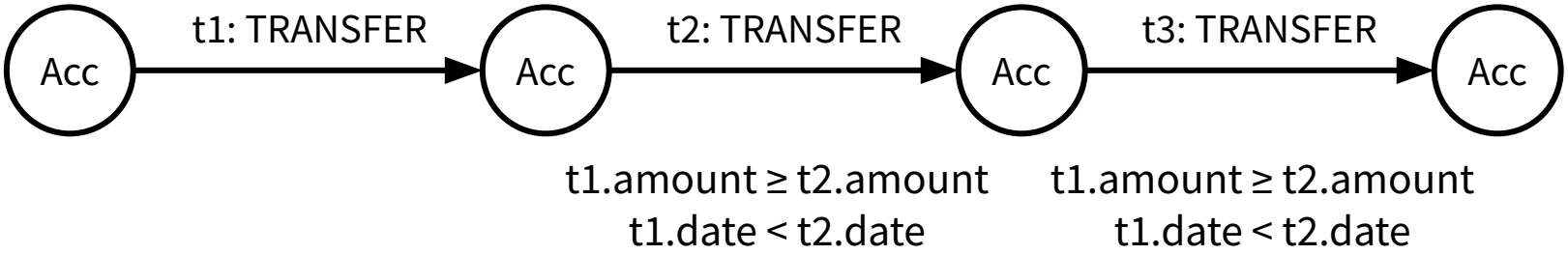
Target: Distributed transactional systems

Financial Benchmark (FinBench)

Originally proposed by the Ant Group, developed with Create Link, Ultipa, etc.

Features:

- Strict latency requirements (P99 < 100 ms), relaxed consistency guarantees
- Truncation (sampling) on more recent edges
- Interesting queries, e.g. REM path queries (Regular Expression with Memory)

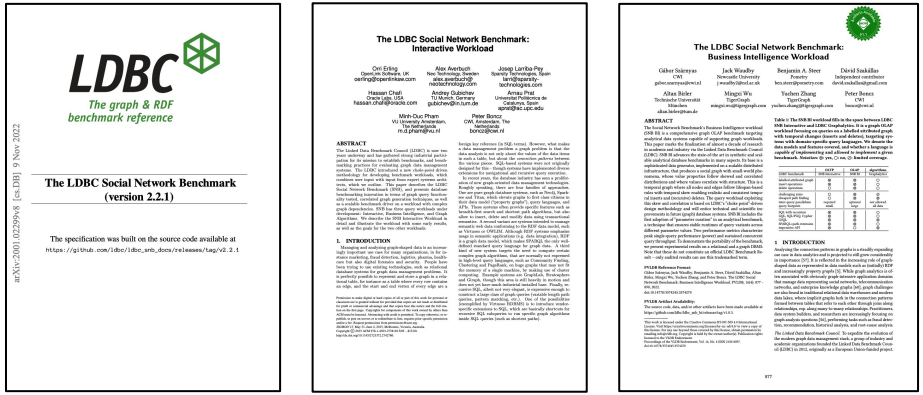


Benchmarking and auditing

Making benchmarks easy to use

For each workload:

- Specification
- Academic paper
- Data generator
- Pre-generated data sets
- Benchmark driver
- 2+ reference implementations



Guidelines:

- How to execute the benchmark correctly
- Validate the results
- Verify ACID-compliance

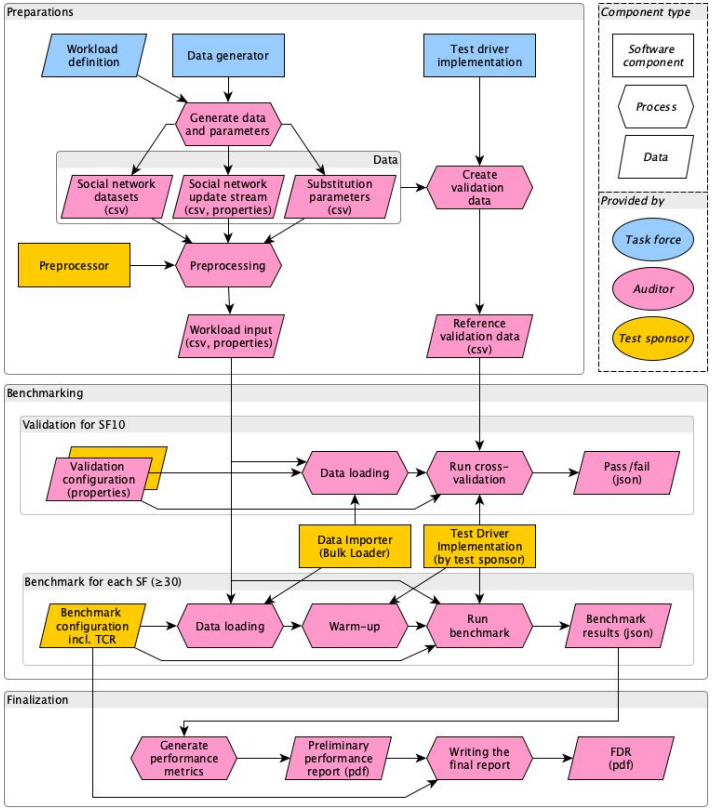
Auditing and trademark

Auditing process:

- Auditors are trained by the LDBC task forces and they take an *auditor exam* to get certified.
- Audits typically cost around 20-50k USD (plus infra costs) and take multiple weeks.

Trademark:

- LDBC is **trademarked** worldwide. Only a **result produced by a certified auditor is an "LDBC benchmark result"**
- Unofficial benchmark results must come with a disclaimer: "This is NOT an official LDBC benchmark result"



LDBC's working groups: graph schema and query languages

Modern graph query languages

Cypher

GSQL

SPARQL

DQL

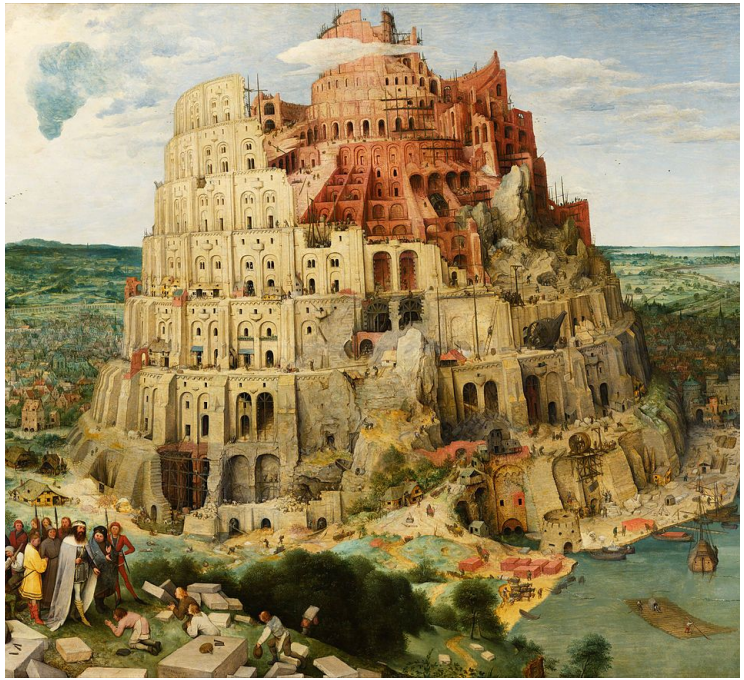
AQL

TypeQL

Gremlin

nGQL

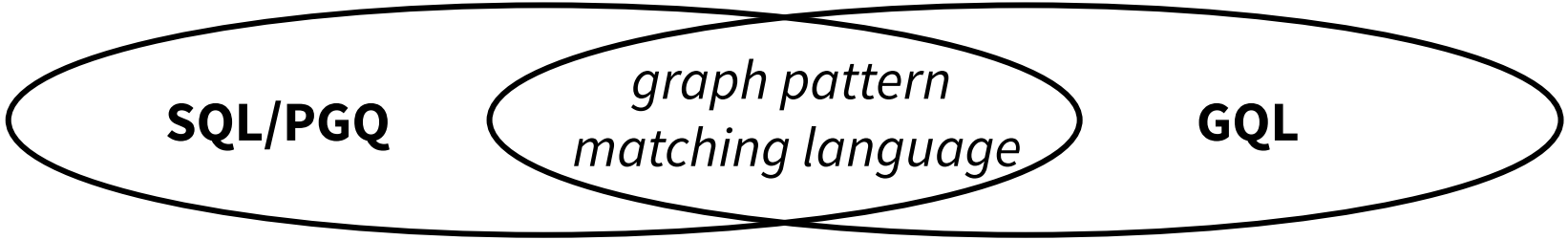
Datalog



LDBC benchmarks define queries in plain text

New ISO standard query languages

- **SQL/PGQ** (Property Graph Queries), part of SQL:2023
- **GQL** (Graph Query Language), to be released in 2024

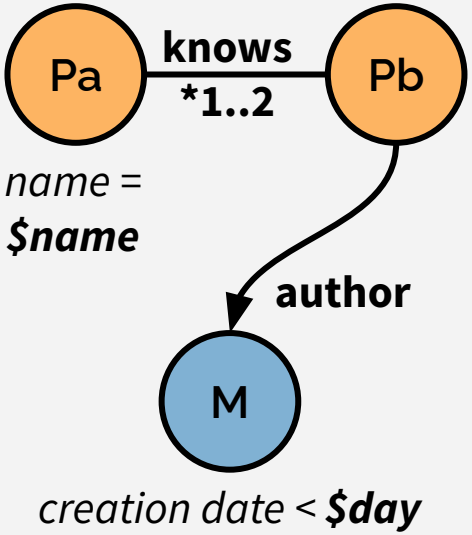


- LDBC has a **liaison with ISO** which allows its members to access to the standard drafts

SQL:1992

```
SELECT DISTINCT m.id
FROM (
 SELECT k.p2id AS id
 FROM person Pa,
      knows k
 WHERE Pa.name = $name
   AND Pa.id = k.p1id
 UNION
 SELECT k2.p2id AS id
 FROM person Pa,
      knows k1,
      knows k2
 WHERE Pa.name = $name
   AND Pa.id = k1.p1id
   AND k1.p2id = k2.p1id
   AND k1.p1id <> k2.p2id
 ) Pb,
 Message m
WHERE Pb.id = m.authorId
  AND m.creationDate < $day
```

Q9(*$name*, *$day*)



SQL/PGQ (SQL:2023)

```
SELECT id
FROM GRAPH_TABLE (socialNetwork
 MATCH ANY ACYCLIC
   (Pa:Person WHERE Pa.name = $name)
   -[:knows]-{1,2} (Pb:Person)
   -[:author]-> (m:Message)
   WHERE m.creationDate < $day
   COLUMNS (m.id))
```

Graph pattern matching language with visual graph syntax inspired by Cypher

GQL

```
MATCH ANY ACYCLIC
   (Pa:Person WHERE Pa.name = $name)
   -[:knows]-{1,2} (Pb:Person)
   -[:author]-> (m:Message)
WHERE m.creationDate < $day
RETURN DISTINCT m.id
```

Q13($src, $dst)

SQL/PGQ (SQL:2023)

```
SELECT length FROM GRAPH_TABLE (sn
 MATCH p = ANY SHORTEST
  (Pa:Person WHERE Pa.name = $src)-[:knows]-*
  (Pb:Person WHERE Pb.name = $dst)
COLUMNS (path_length(p) AS length))
```

SQL:1999

```
WITH RECURSIVE ps(sp, ep, path, eR) AS (
  SELECT p1id AS sp, p2id AS ep, [p1id, p2id] AS path, (p2id = $dst) AS eR
    FROM knows WHERE sp = $src UNION ALL SELECT ps.sp AS sp, p2id AS ep,
    array_append(path, p2id) AS path, max(CASE WHEN p2id = $dst THEN 1 ELSE 0 END)
    OVER (ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING) AS eR
    FROM ps JOIN knows ON ps.ep = p1id WHERE NOT EXISTS
    (SELECT 1 FROM ps pps WHERE list_contains(pps.path, p2id)) AND ps.eR = 0)
SELECT min(length(path)) AS length FROM ps WHERE ep = $dst
```

LDBC working groups

Graph schema: Balancing expressive power, usability and tractability

- PG-Keys: Keys for Property Graphs (SIGMOD’21)
- PG-Schema: Schemas for Property Graphs (SIGMOD’23)

Graph query languages: Formalizing semantics, ensuring tractability

- G-CORE (SIGMOD’18)
- Graph Pattern Matching in GQL and SQL/PGQ (SIGMOD’23)
- GPC: A Pattern Calculus for Property Graphs (PODS’23)

LDBC organization

LDBC organization

LDBC is registered in the UK as a non-profit company

Annual membership fees (approx.):

- sponsors: 11,000 USD
- companies: 2,800 USD
- institutions: 1,400 USD

Approx. 100,000 USD per year revenue

Organizational structure

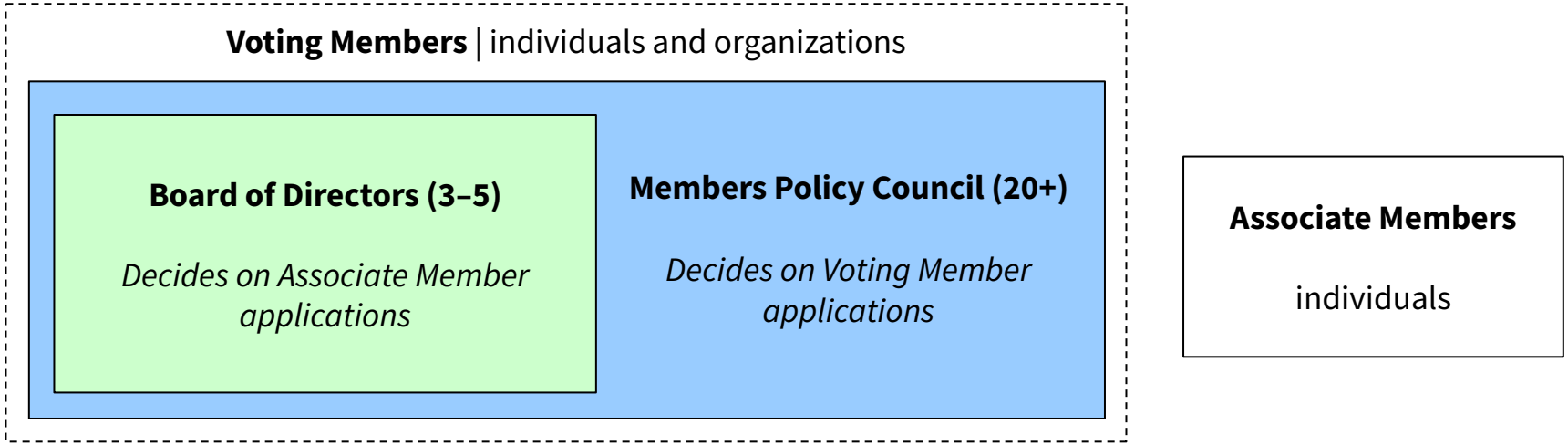
Old structure: member organizations delegate directors to the board

This made the company suspicious

- 100,000 USD per year revenue
- 20+ directors with different nationalities

→ we restructured

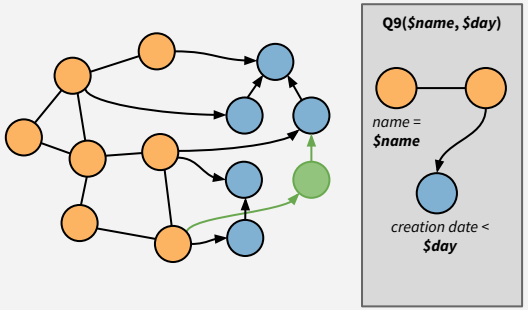
Organizational structure



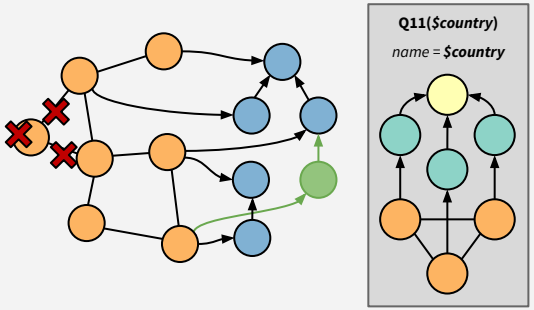
The membership form is 32 pages (patent declaration, CLA, etc.)

Summary

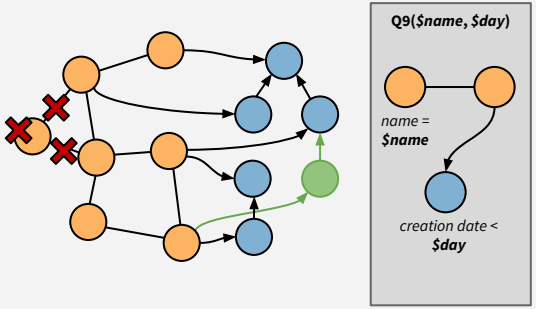
SNB Interactive v1



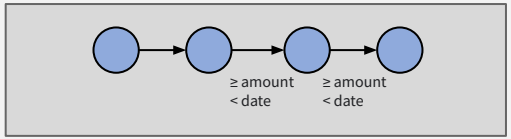
SNB Business Intelligence



SNB Interactive v2



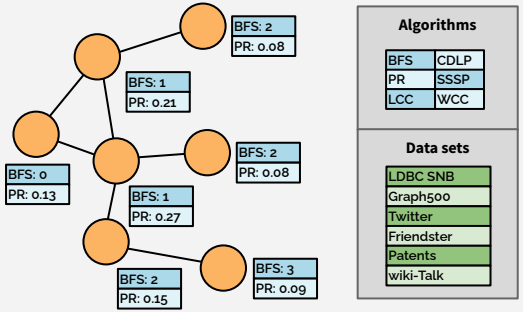
Financial Benchmark



Traversal with truncation

Strict latency bound (P99 < 100 ms)

Graphalytics



Semantic Publishing Benchmark

Target: **RDF/SPARQL**

Domain: **Media/publishing industry**

Inferencing & continuous updates

LDBC's main challenges

- Handling large-scale data sets is expensive:
  - Data generation – SF30k in AWS EMR – 120 instances, TODO hours, TODO USD
  - Data hosting
  - Transferring data
- Audits are complex
  - 8–20 weeks
  - Long and expensive
- Most audited results use imperative languages
  - GQL and SQL/PGQ may help
- Developing a benchmark takes 5+ person-years
  - No standard language, implementations take a long time
  - Hard to obtain a good baseline system

LDBC

The graph & RDF benchmark reference